Clinical correlation: osteoporosis

Rattana leelawattana

Endocrine unit, Dep of Medicine, Fac of Medicine

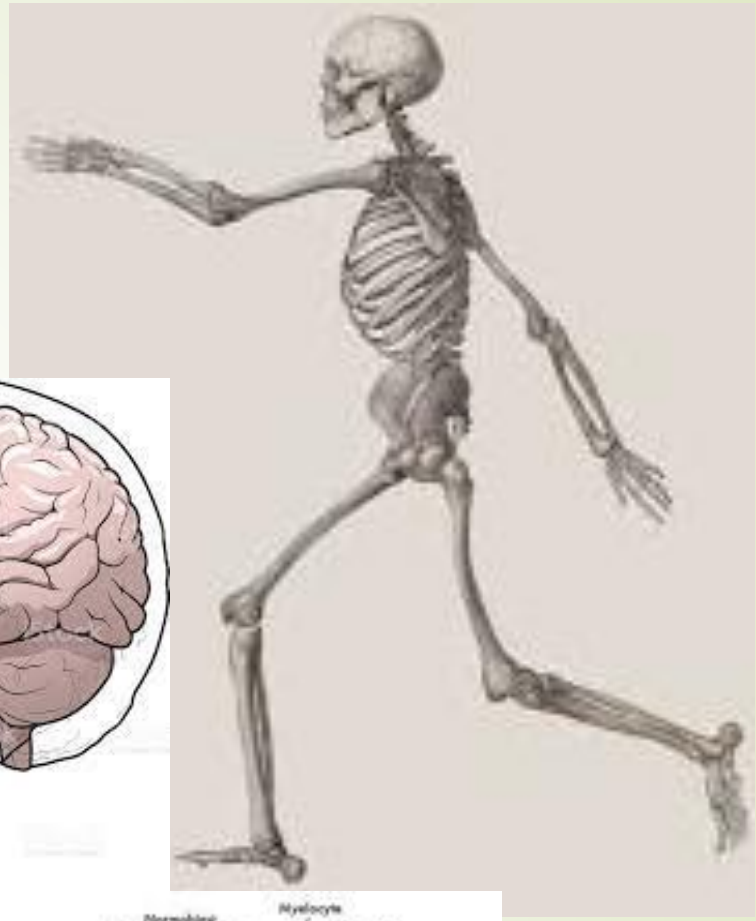
PSU

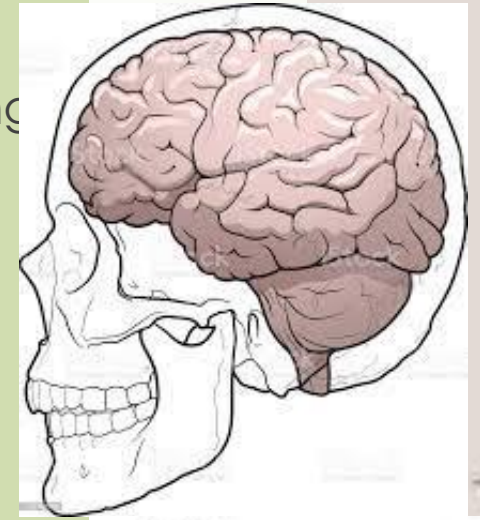
Objectives

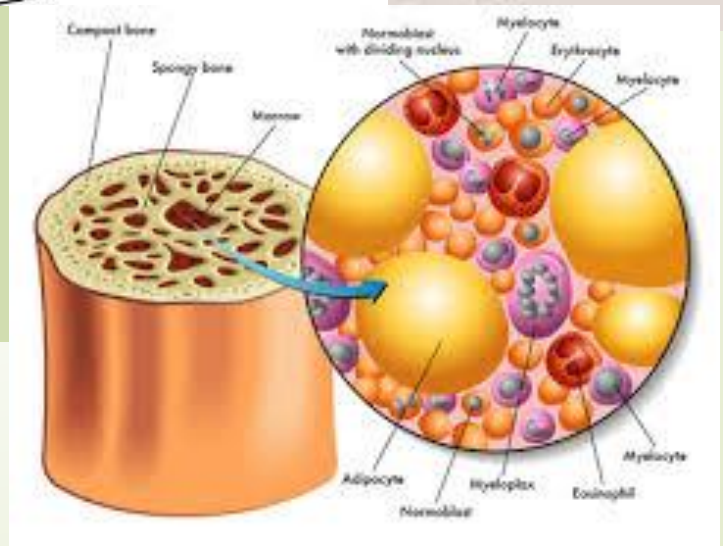
- Bone biology, physiology
- Osteoporosis
  - Definition
  - Diagnosis
  - Fragility fracture
  - reatment

Bone functions

- Support body
- Mobilization: skeleton, chewing, breathing
- Protection
  - Vital structures (brain, cord, lung, heart)
  - House of blood cells
- Metabolic
  - Calcium
  - Acid-base
  - Hormone (FGF23, osteocalcin)







Bone & acid/base modification

- Bone = CO_3 storage house
- Metabolic acidosis
  - physicochemical milieu allowing bone resorption
  - Suppress osteoblast function
  - Osteoclast recruitment, maturation, survival

Bone

- Growing
- Bone marrow space
- Strong
- Renew/repair
- Resorption/release

Trabecular number

Osteoclast

Modeling

Osteoblast

Osteoid

Remodeling

Trabecular

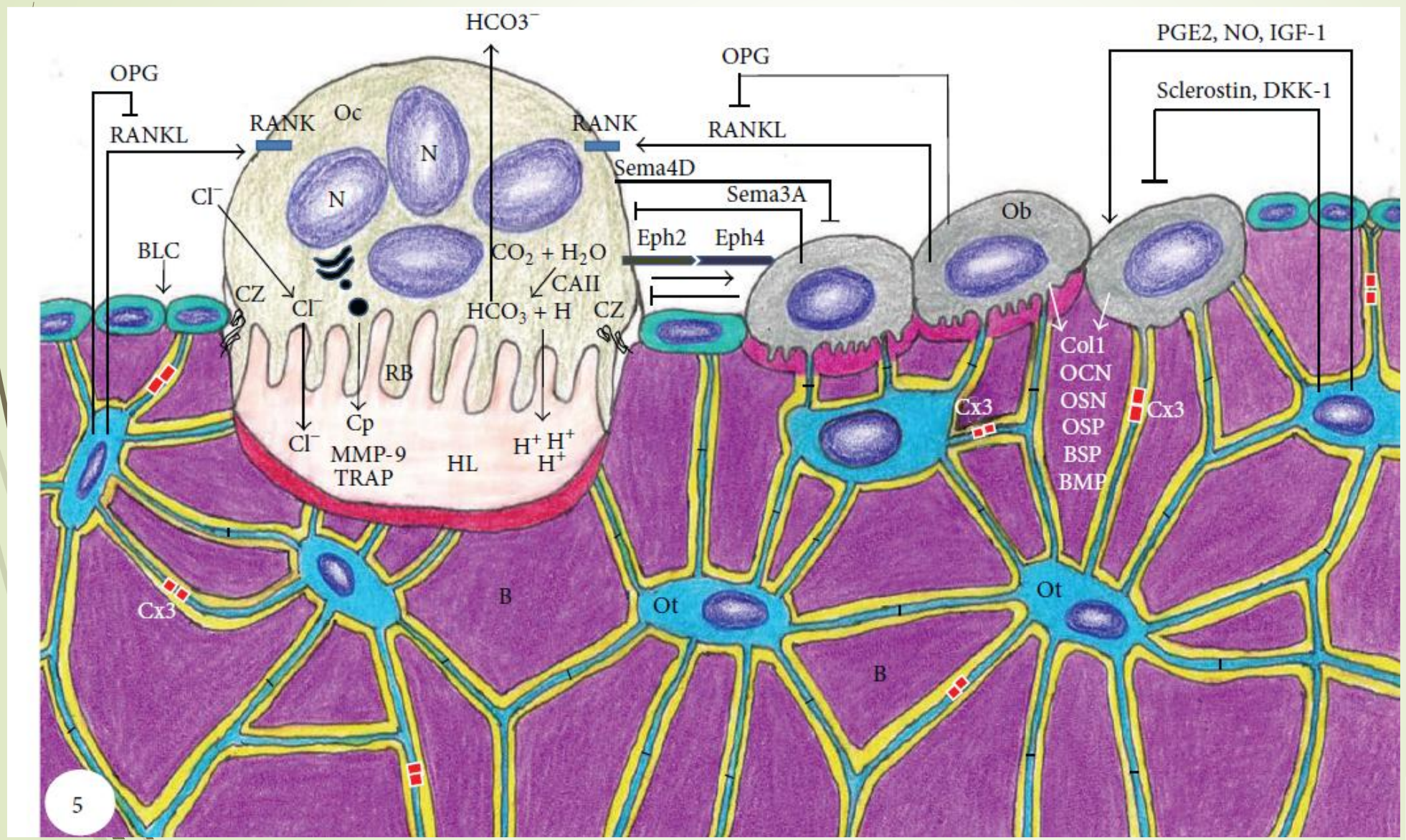
Mineralization

Osteocyte

Cortical porosity

Bone resorption

Bone



BioMed Research International
Volume 2015, Article ID 421746,

Bone cells

Osteoblast

- Mesenchymal stem cell (bone, muscle, fat, fibroblast)
- Bone forming (collagen synthesis)
- PTH, estrogen receptor
- Cuboidal shape
- Alkaline phosphatase

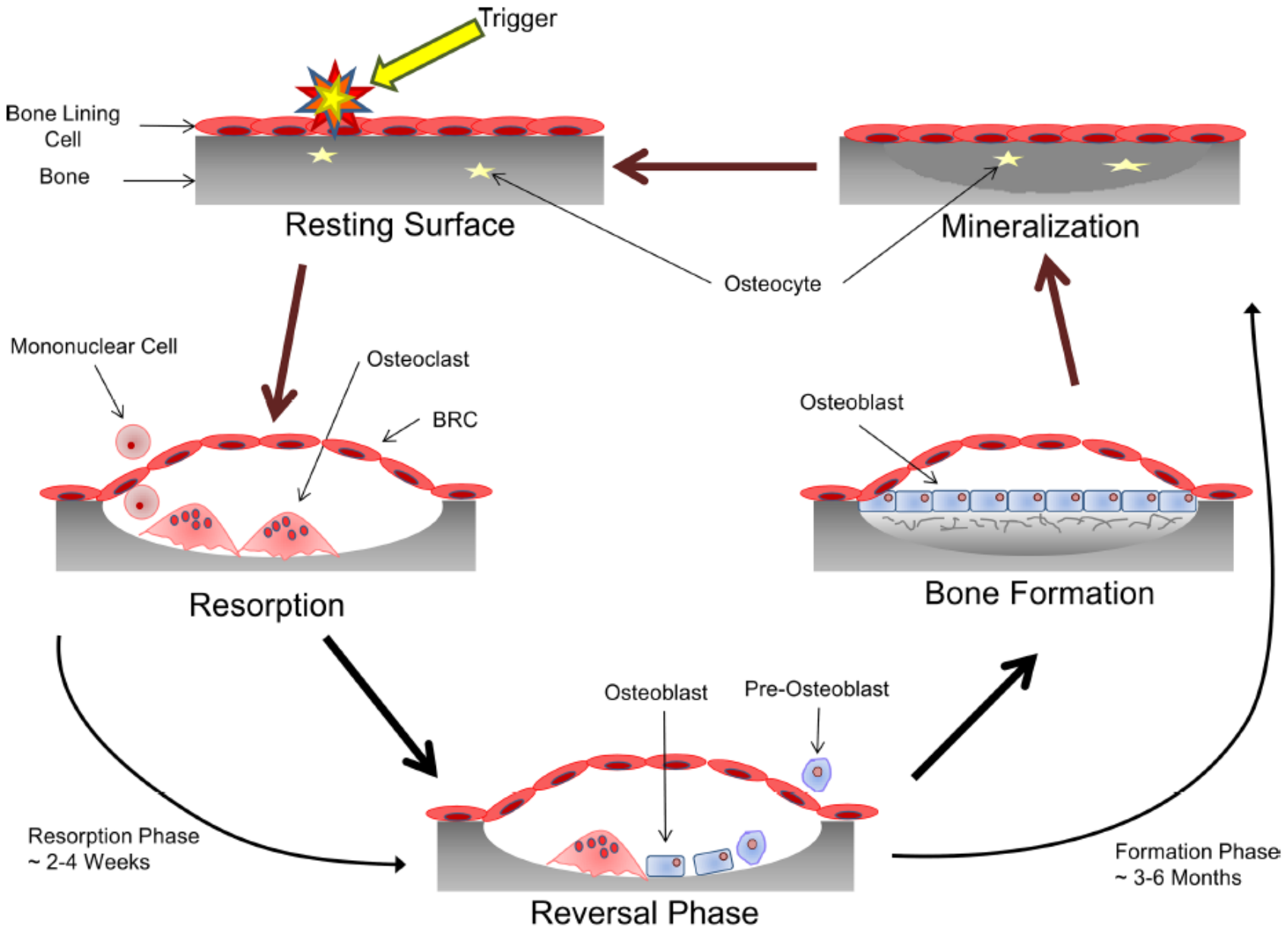
Osteocyte

Trapped osteoblast
Mechanosensor
Control bone turnover
Release FGF23, osteocalcin, sclerostin, RANKL

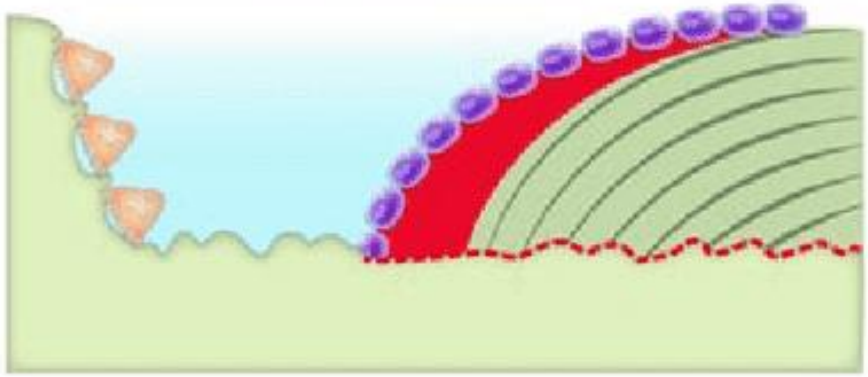
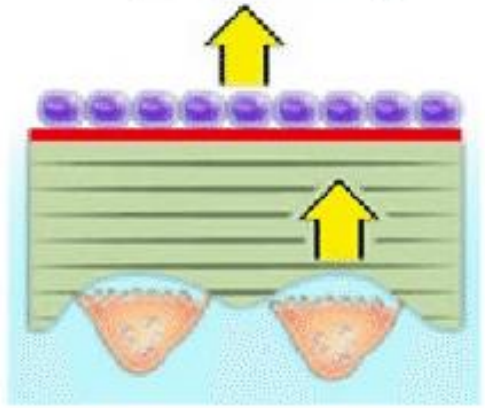
Osteoclast

- Macrophage
- Bone resorption
- Multinucleated giant cell
- Calcitonin, TNF receptor
- Acid phosphatase
- Cathepsin K & many lysozymes
- Acid secretion

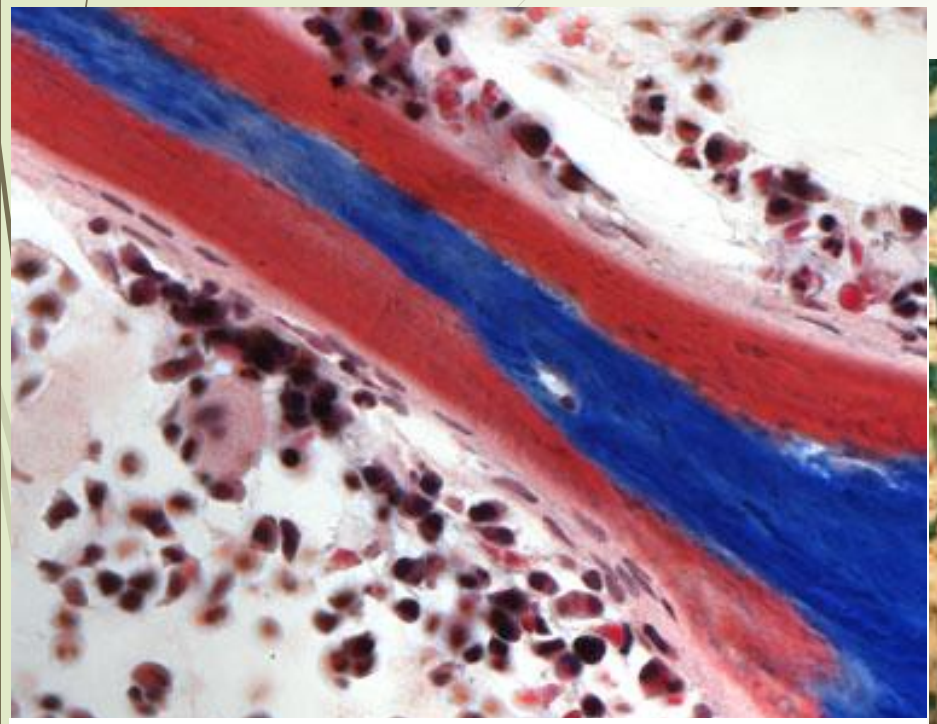
Bone remodeling



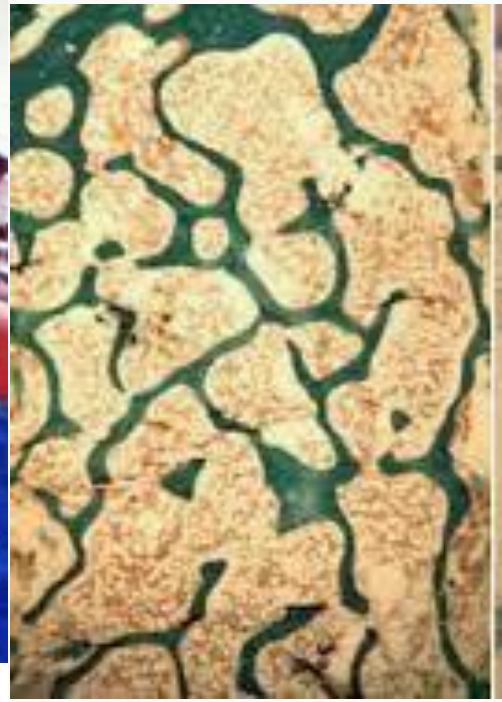
Bone modeling

	Remodeling	Modeling
		
Arrangement of Osteoclasts& Osteoblasts:	Same surface	Different surfaces
Activity:	Cyclical	Continuous
Direct Coupling:	Maybe	No
Effect on Bone Mass:	Slow	Fast

What condition is modeling found



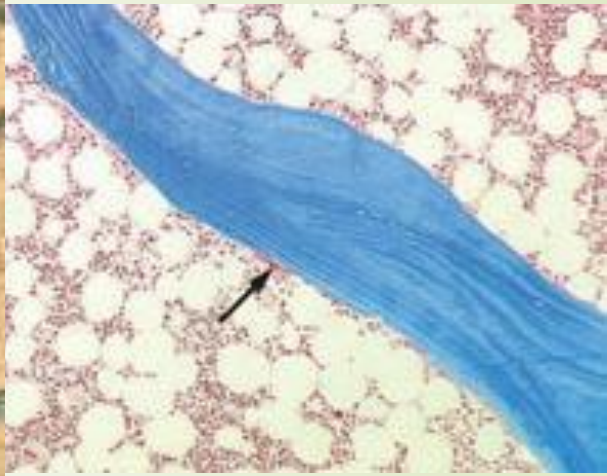
Osteomalacia: mineralized bone = blue, matrix = red



Normal

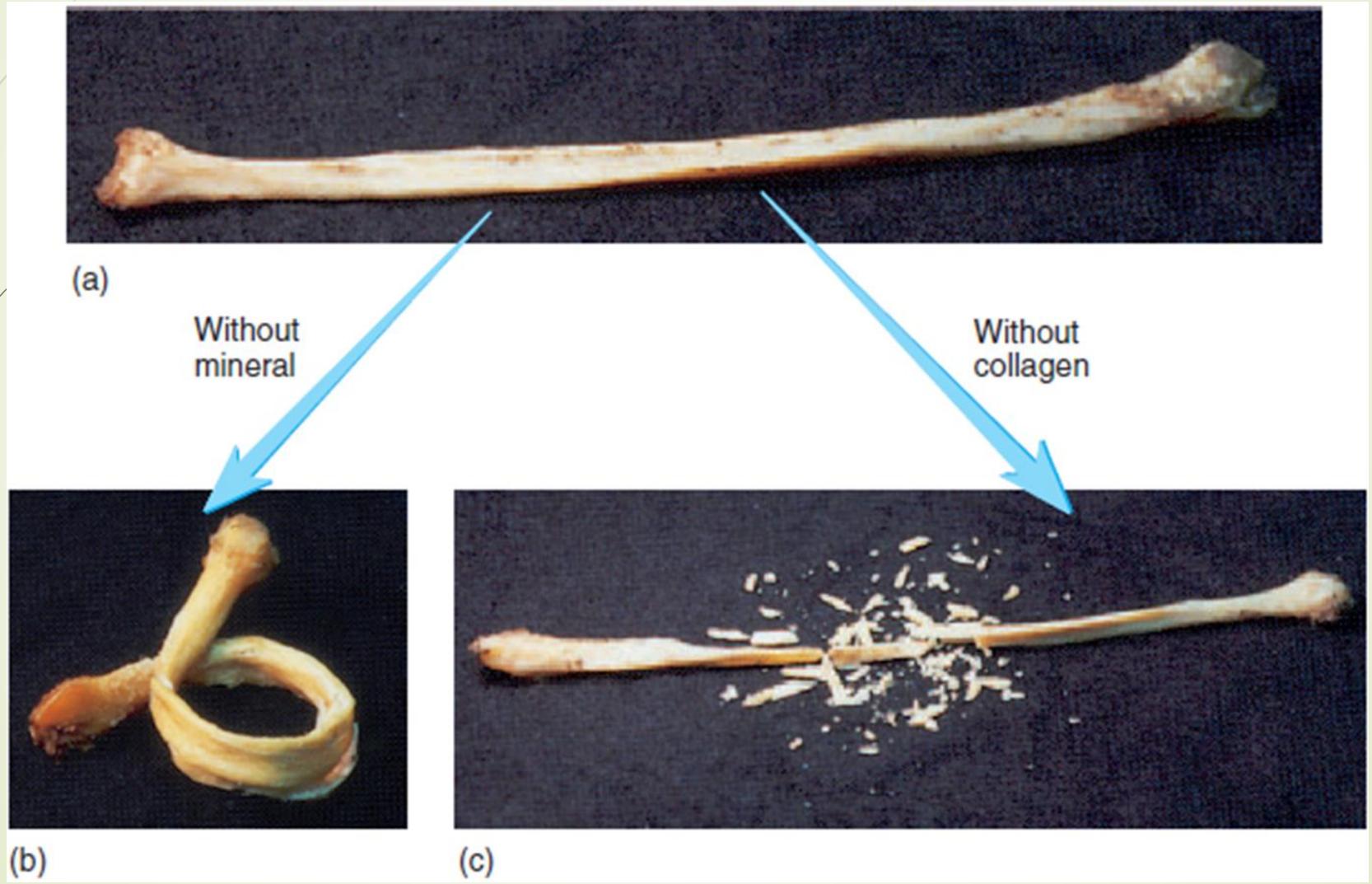


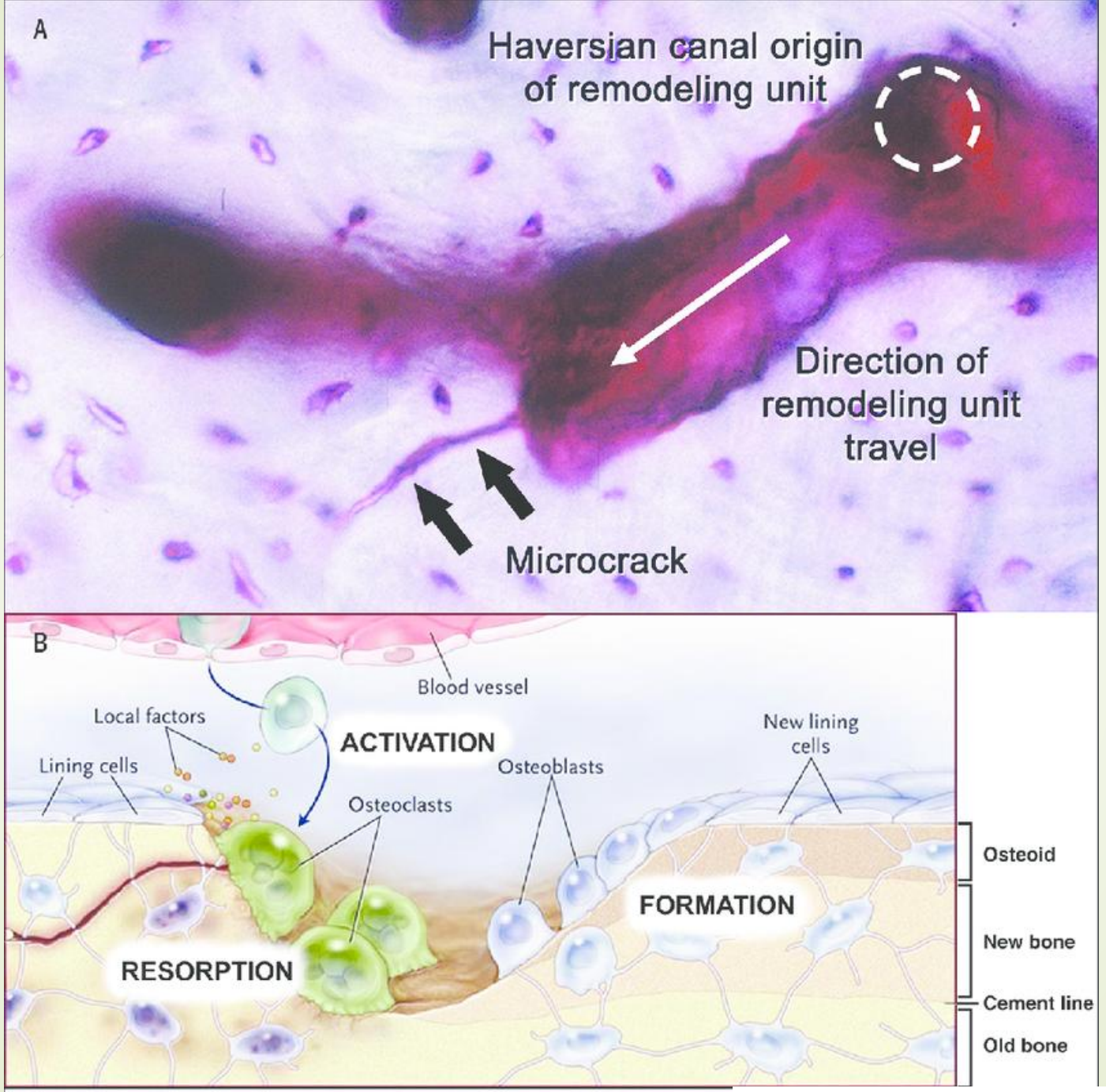
osteoporosis



Adynamic bone
Blue = mineralized bone, matrix = red

Bone = matrix + mineral

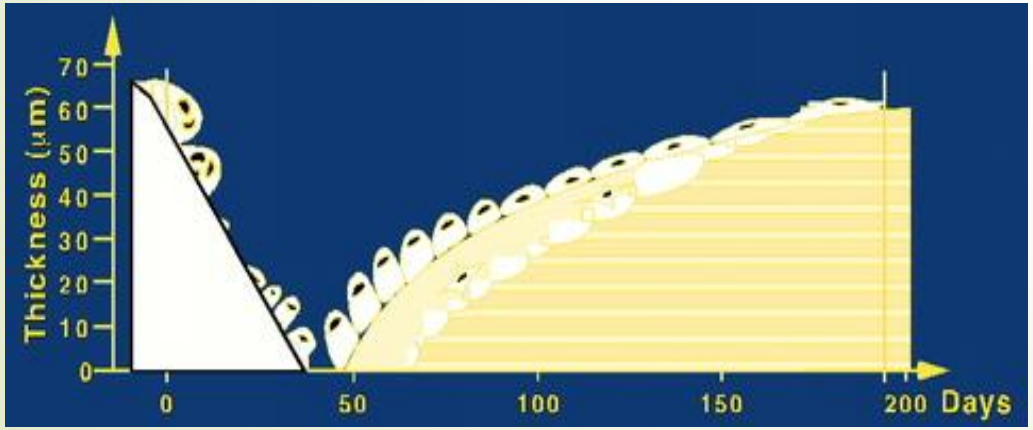


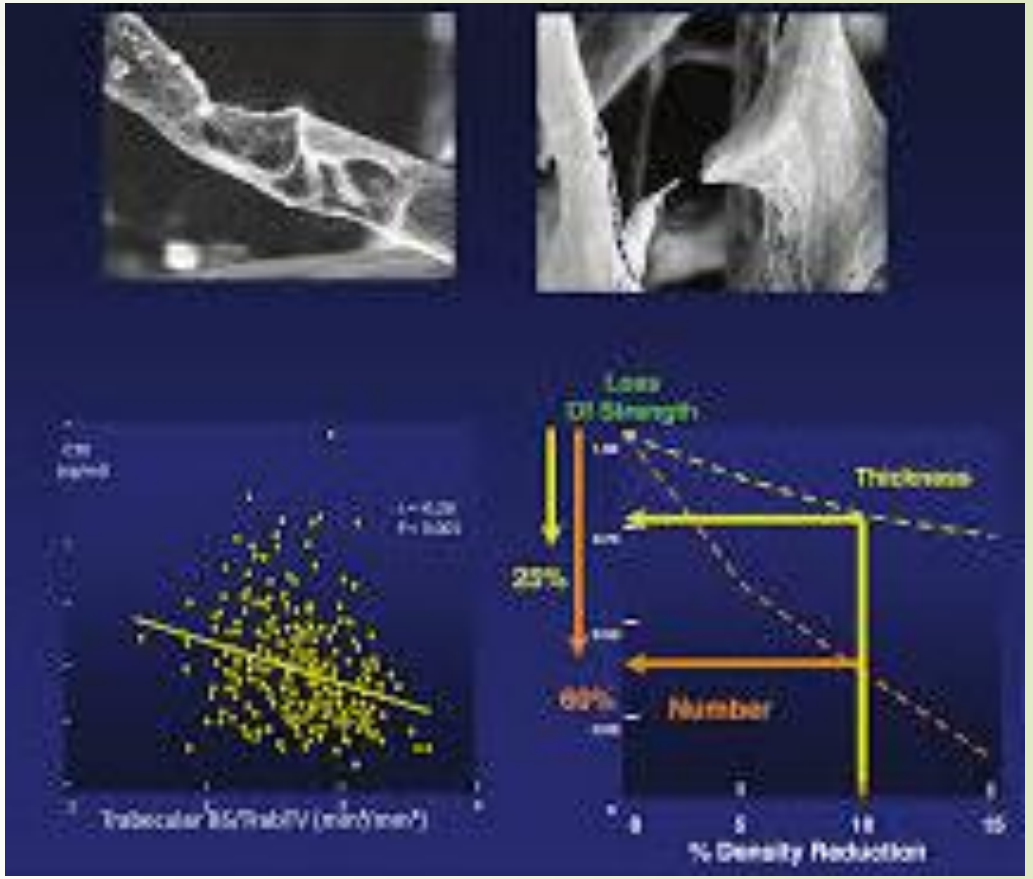


J Orthop Sports Phys Ther 2014;44(10):749-765.

Bone turnover

- Keep new bone
- Osteocyte, osteoblast, osteoclast
- Too much vs too little
- Couple





Bone

- Growing
- Bone marrow space
- Strong
- Renew/repair
- Resorption/release

Trabecular number

Osteoclast

Modeling

Osteoblast

Osteoid

Remodeling

Trabecular

Mineralization

Osteocyte

Cortical porosity

Bone resorption

Bone

- Growing
- Bone marrow space
- Strong
- Renew/repair
- Resorption/release

Trabecular number

Osteoclast

Modeling

Osteoblast

Osteoid

Remodeling

Trabecular

Mineralization

Osteocyte

Cortical porosity

Bone

- Growing
- Bone marrow space
- Strong
- Renew/repair
- Resorption/release

Trabecular number

Osteoclast

Modeling

Osteoblast

Osteoid

Remodeling

Trabecular

Mineralization

Osteocyte

Cortical porosity

Bone resorption

Bone

- Growing
- Bone marrow space
- Strong
- Renew/repair
- Resorption/release

Trabecular number

Osteoclast

Modeling

Osteoblast

Osteoid

Remodeling

Trabecular

Mineralization

Osteocyte

Cortical porosity

Bone resorption

Bone

- Growing
- Bone marrow space
- Strong
- Renew/repair
- Resorption/release

Trabecular number

Osteoclast

Modeling

Osteoblast

Osteoid

Remodeling

Trabecular

Mineralization

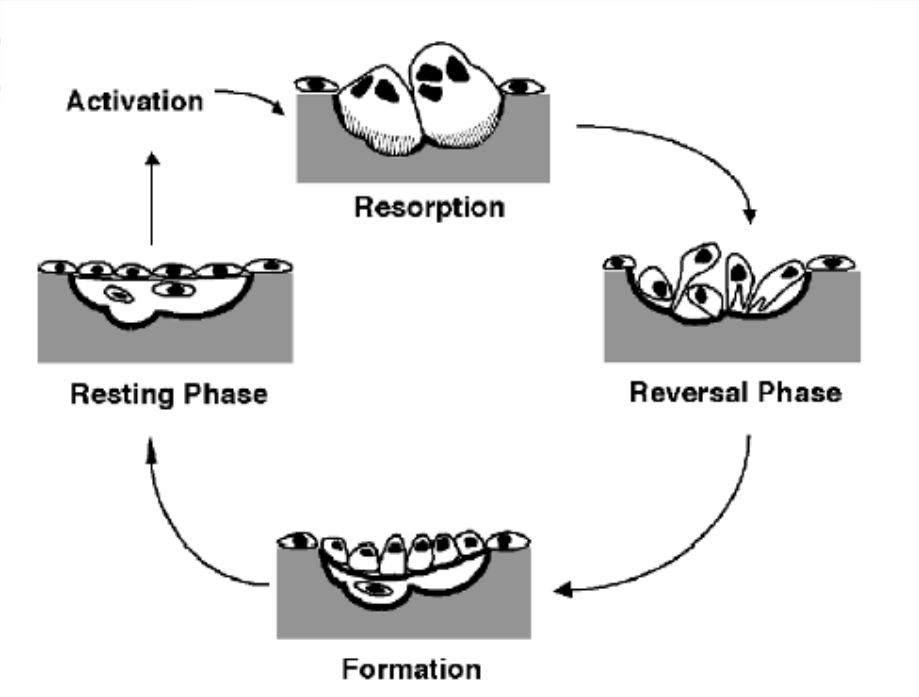
Osteocyte

Cortical porosity

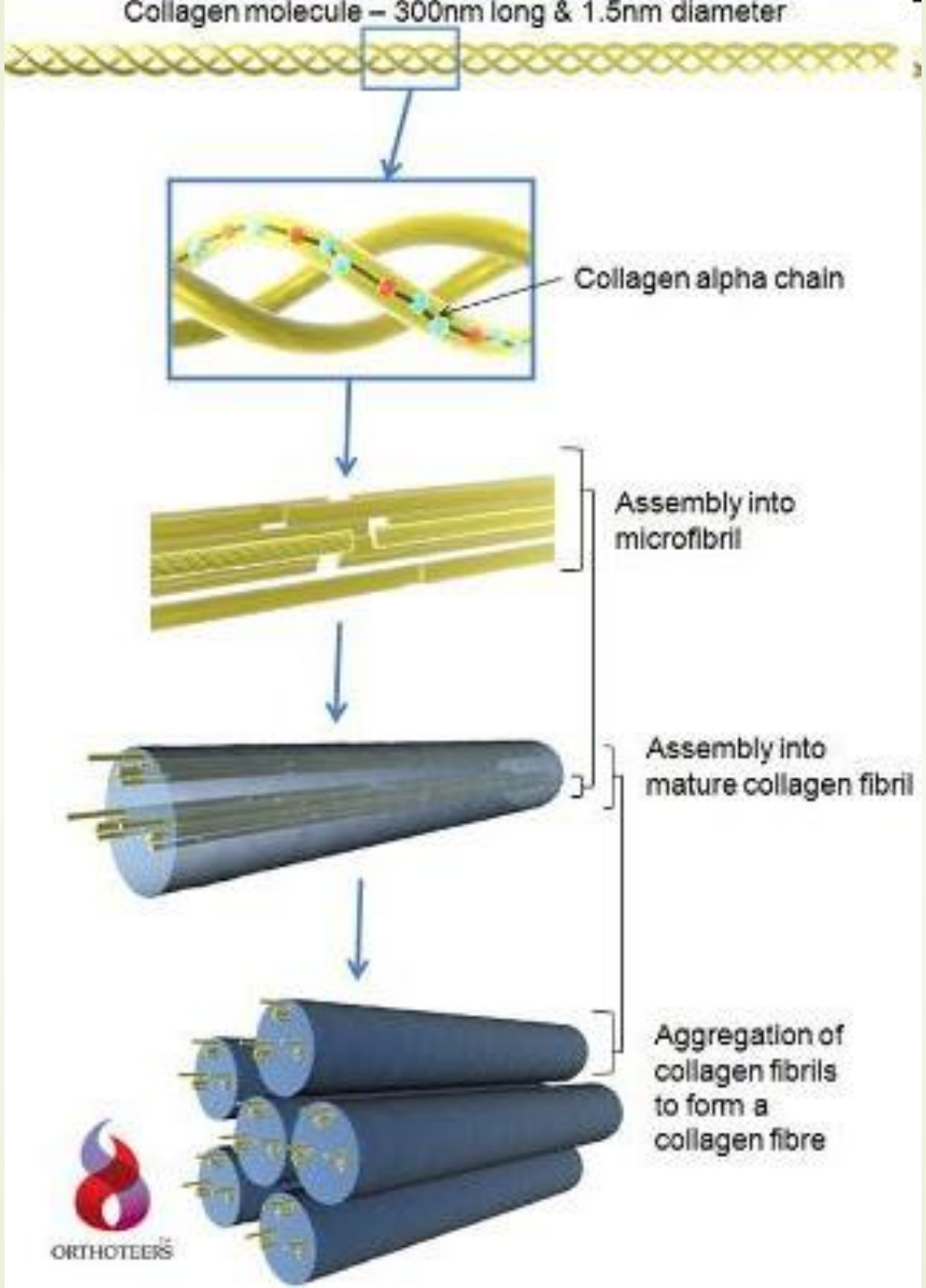
Bone = matrix + mineral

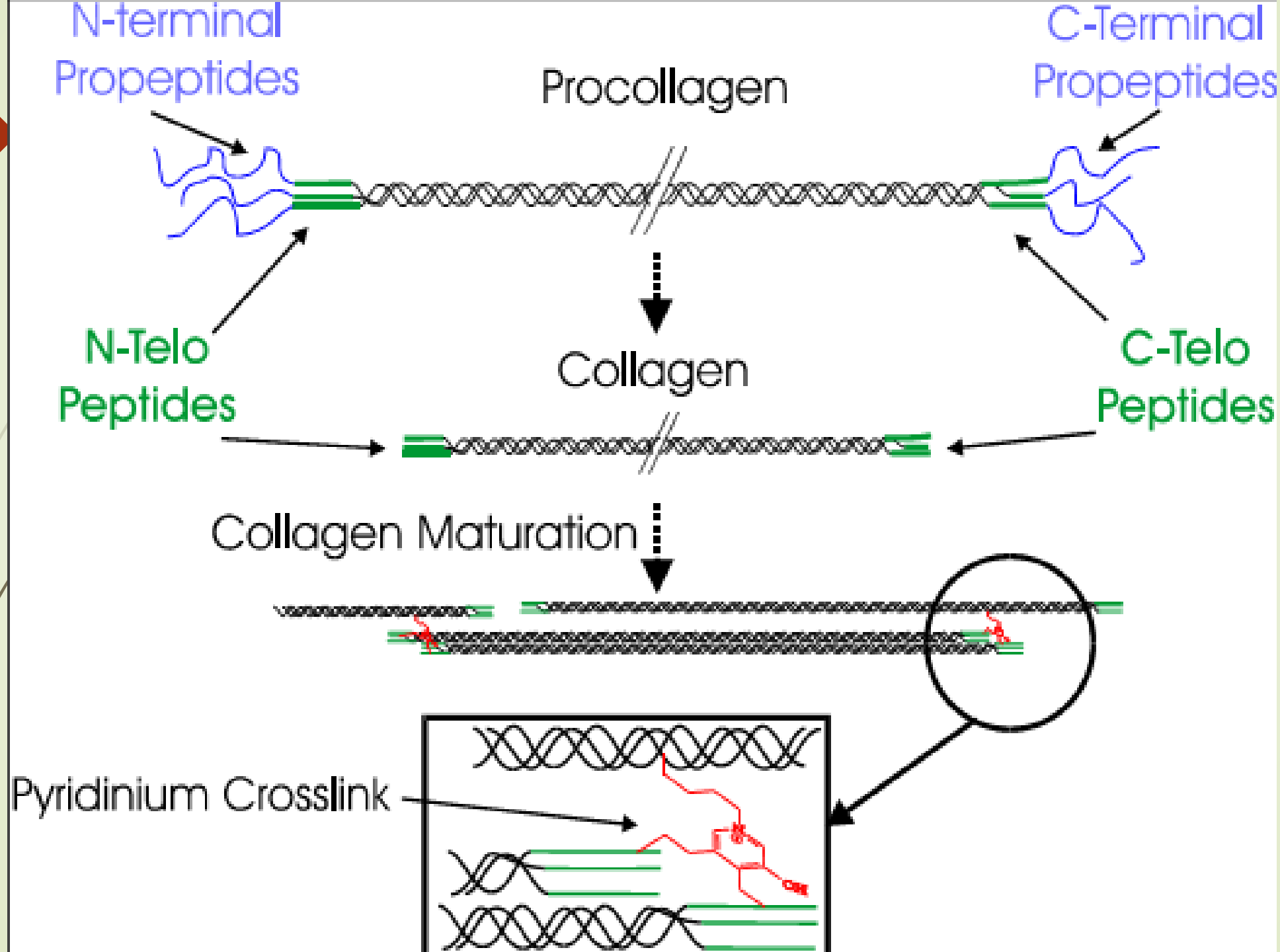
- Mature bone matrix is approximately 35% organic and 65% inorganic material
- Organic: collagen (90%) and non-collagenous proteins (10%: osteocalcin, osteonectin, osteopontin and bone sialoprotein)
- Matrix harden by mineral, then entrap cell

Wh



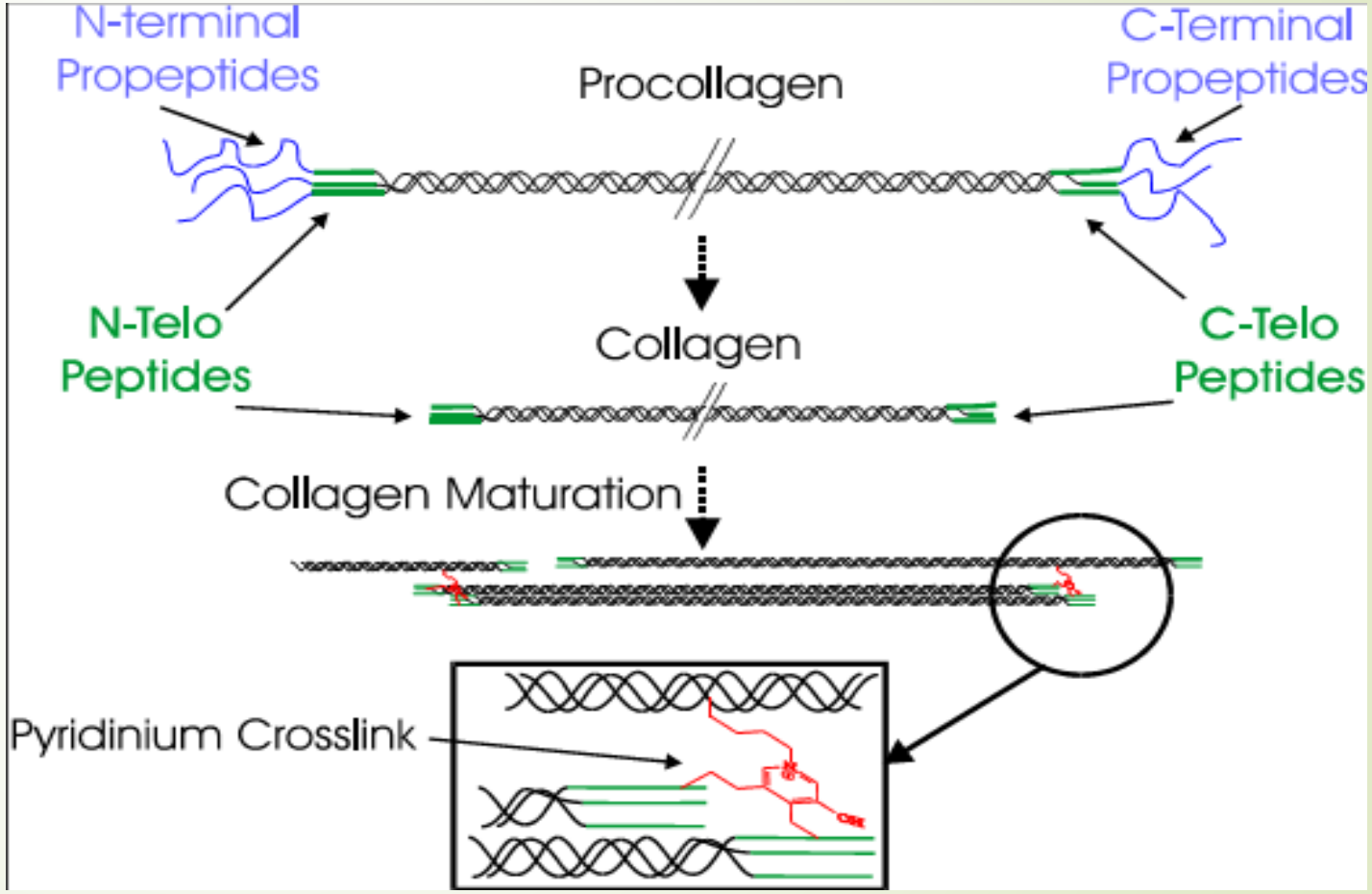
- Formation
  - sP1NP
  - sP1CP
  - sAlp: bone, others
  - sOsteocalcin
- Resorption
  - sCTX
  - sNTX
  - sTRAP5b
  - U Deoxypyridinoline
  - U Pyridinoline
  - U Hydroxyproline

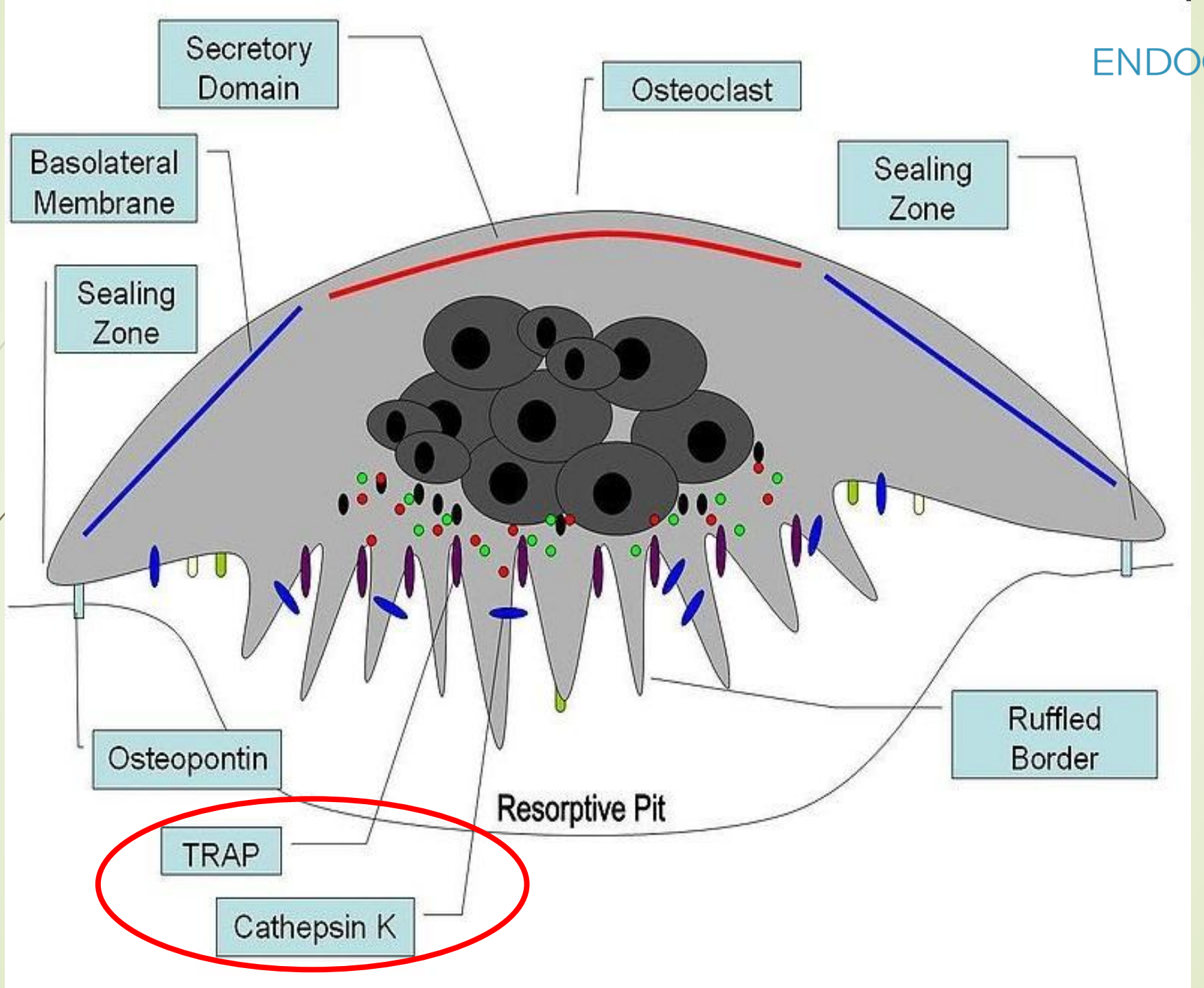

Collagen molecule – 300nm long & 1.5nm diameter
Collagen alpha chain
Assembly into microfibril
Assembly into mature collagen fibril
Aggregation of collagen fibrils to form a collagen fibre
ORTHOTEERS


N-terminal
Propeptides
Procollagen
C-Terminal
Propeptides
N-Telo
Peptides
Collagen
C-Telo
Peptides
Collagen Maturation
Pyridinium Crosslink

Bone marker

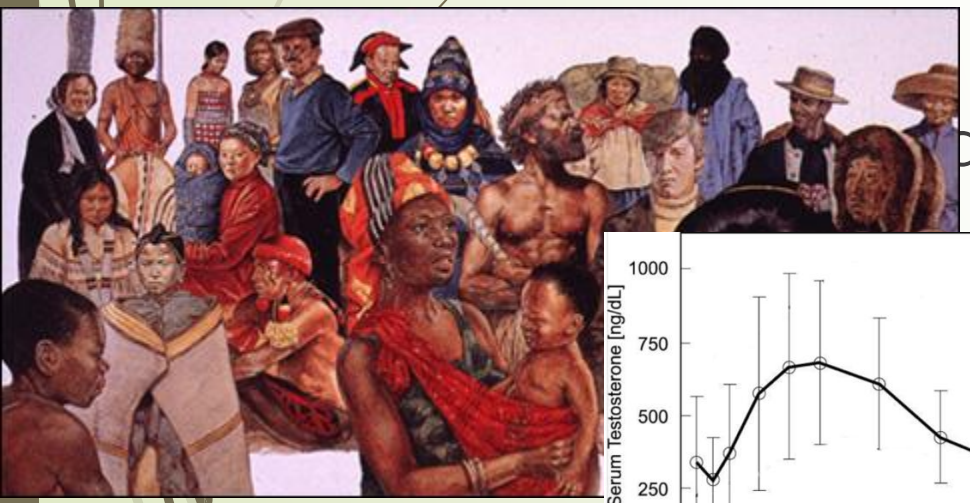
- P1NP = Procollagen type1 amino terminal peptide
- P1CP = Procollagen type1 carboxy terminal peptide
- CTX = C terminal telopeptide crosslink
- NTX = N terminal telopeptide crosslink




Secretory Domain
Osteoclast
Basolateral Membrane
Sealing Zone
Sealing Zone
Ruffled Border
Osteopontin
Resorptive Pit
TRAP
Cathepsin K

Variability

- Specificity of “formation” vs “resorption”, bone vs nonbone
- Pre-analytic variable: biologic variable, sam



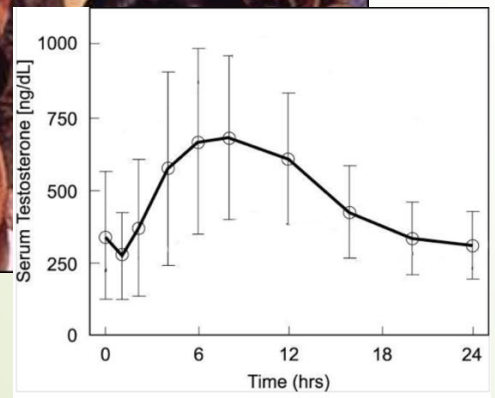
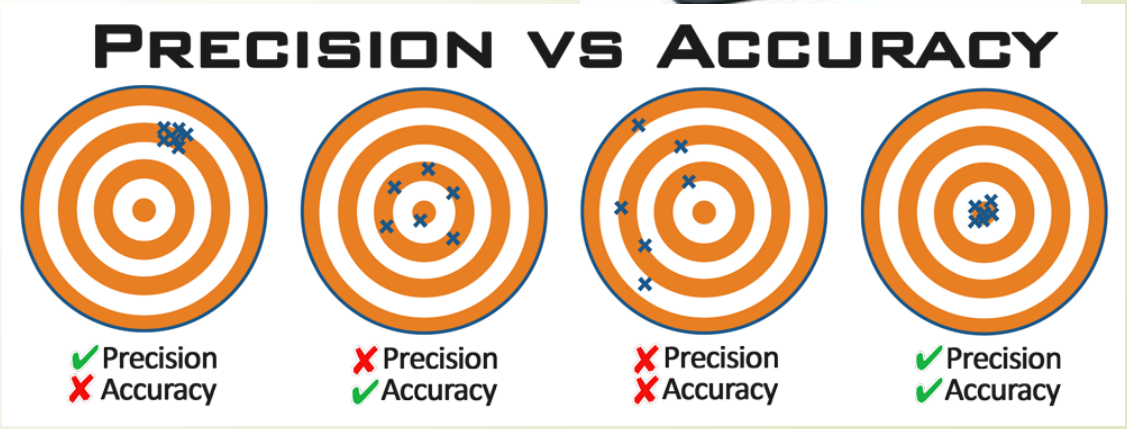


Figure 1. Diurnal variation of serum testosterone levels each day. The levels are highest in the early morning, dropping progressively during the day.

ariable: te e, machine

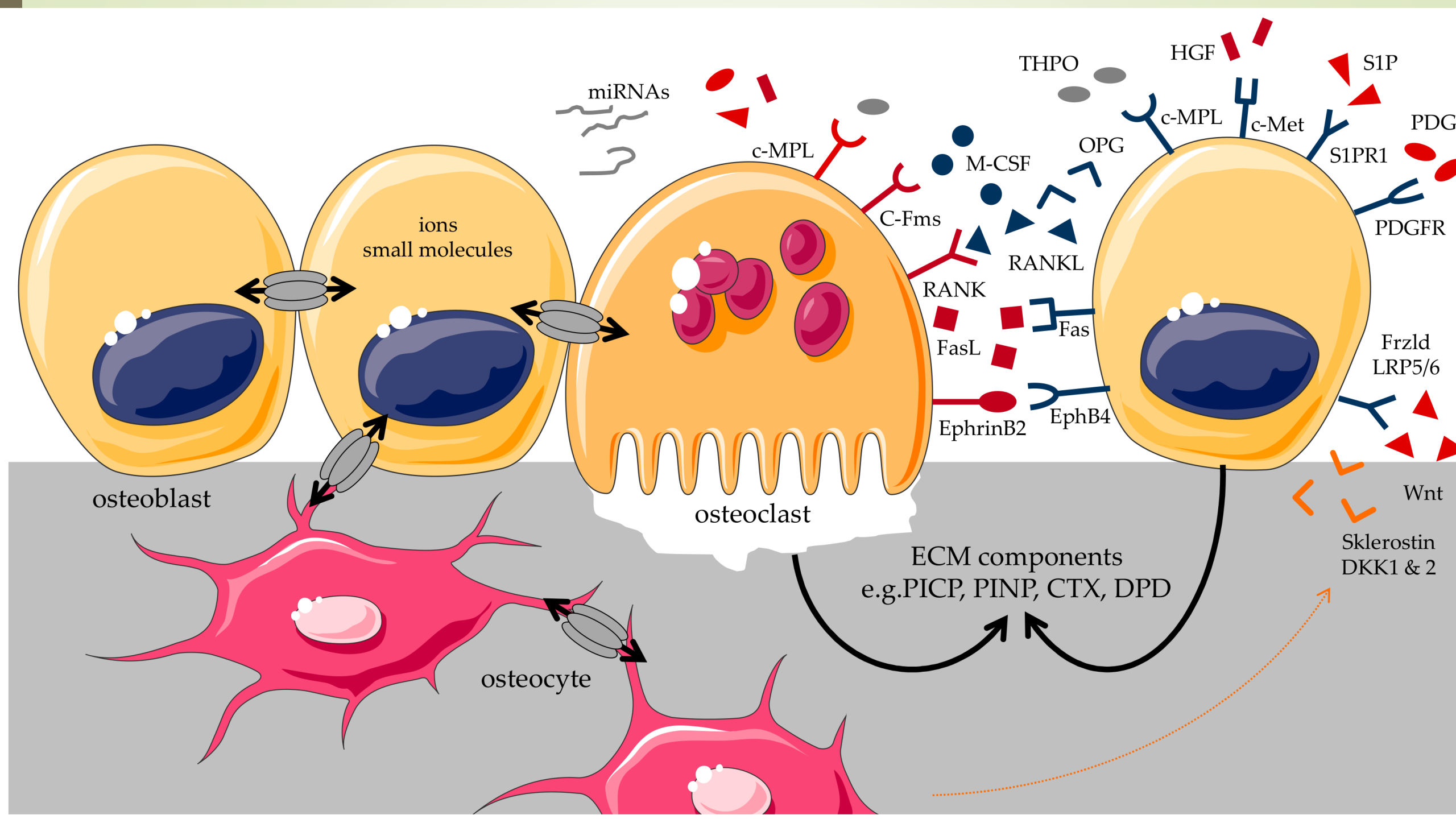


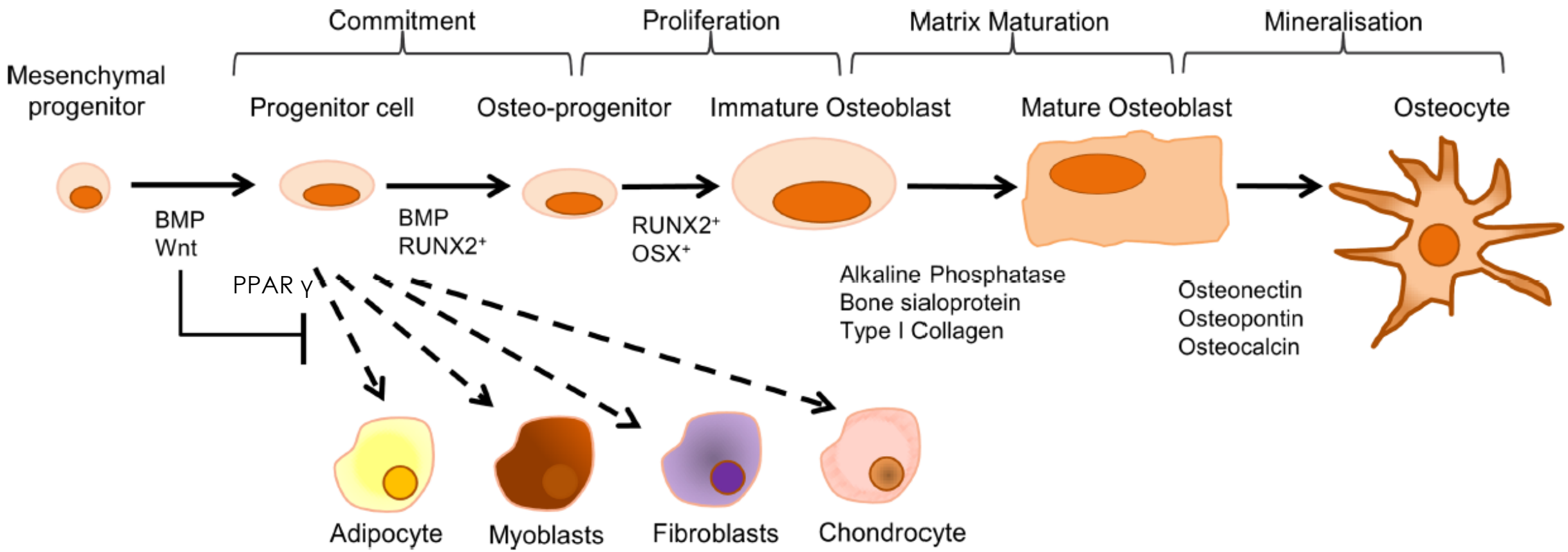


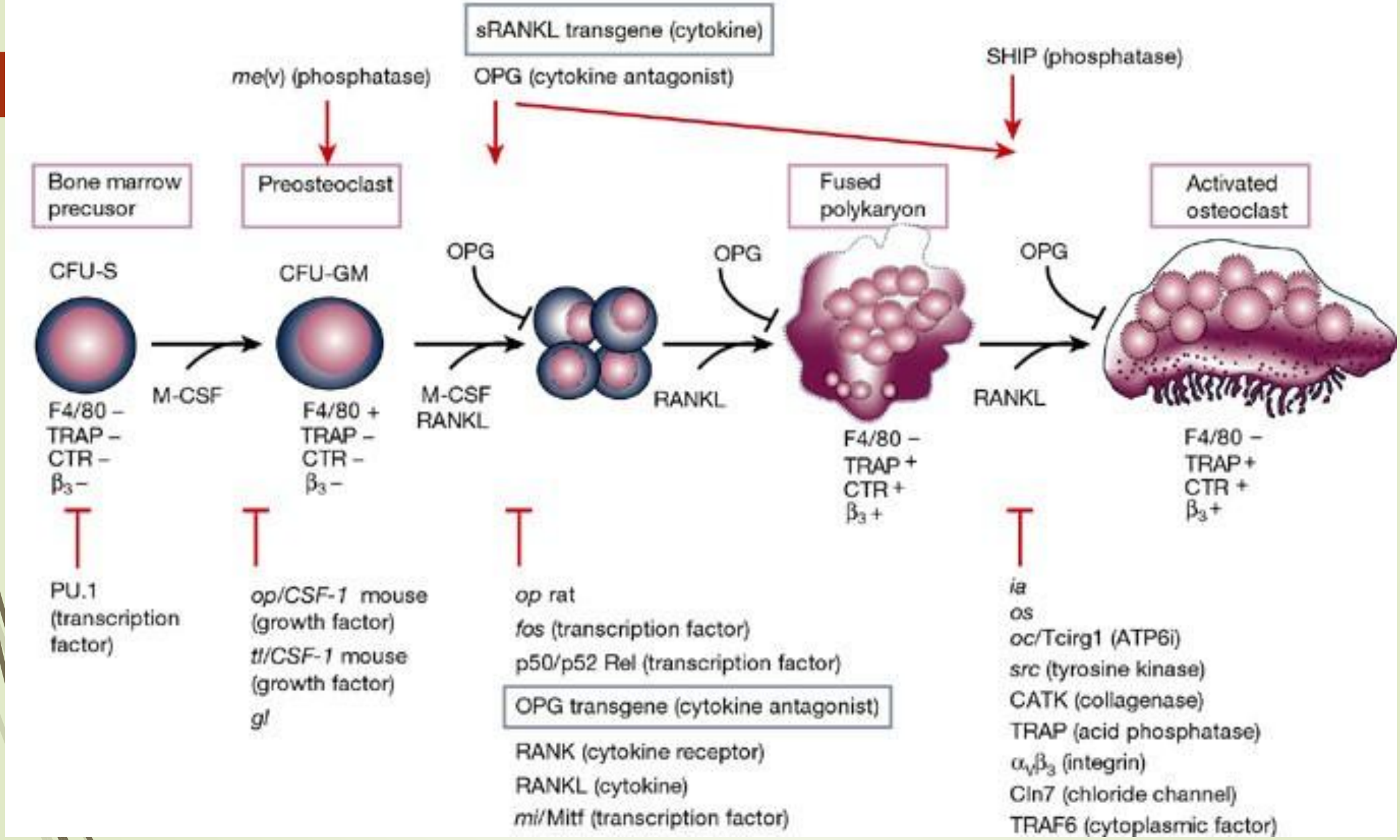


Mineral of bone

	Range in Literature for Healthy Adult Whole Bone (wt% of ash)
Calcium	**32.6-39.5**
Magnesium	0.32-0.78
Sodium	0.26-0.82
Phosphorus	**13.1-18.0**
Carbonate	3.2-13
Citrate	0.04-2.67
Fluoride	0.02-0.207


miRNAs
c-MPL
C-Fms
M-CSF
RANK
RANKL
FasL
Fas
EphrinB2
EphB4
THPO
HGF
S1P
c-MPL
c-Met
OPG
S1PR1
PDG
PDGFR
Frzld
LRP5/6
Wnt
Sklerostin
DKK1 & 2
ions
small molecules
osteoblast
osteoclast
osteocyte
ECM components
e.g.PICP, PINP, CTX, DPD


Commitment
Proliferation
Matrix Maturation
Mineralisation
Mesenchymal progenitor
Progenitor cell
Osteo-progenitor
Immature Osteoblast
Mature Osteoblast
Osteocyte
BMP
Wnt
BMP
RUNX2+
RUNX2+
OSX+
PPAR γ
Alkaline Phosphatase
Bone sialoprotein
Type I Collagen
Osteonectin
Osteopontin
Osteocalcin
Adipocyte
Myoblasts
Fibroblasts
Chondrocyte


sRANKL transgene (cytokine)
me(v) (phosphatase)
OPG (cytokine antagonist)
SHIP (phosphatase)
Bone marrow precusor
Preosteoclast
Fused polykaryon
Activated osteoclast
CFU-S
CFU-GM
OPG
OPG
OPG
M-CSF
M-CSF
RANKL
RANKL
RANKL
F4/80 –
TRAP –
CTR –
β3 –
F4/80 +
TRAP –
CTR –
β3 –
F4/80 –
TRAP +
CTR +
β3 +
F4/80 –
TRAP +
CTR +
β3 +
PU.1
(transcription factor)
op/CSF-1 mouse
(growth factor)
tl/CSF-1 mouse
(growth factor)
gl
op rat
fos (transcription factor)
p50/p52 Rel (transcription factor)
OPG transgene (cytokine antagonist)
RANK (cytokine receptor)
RANKL (cytokine)
mi/Mitf (transcription factor)
ia
os
oc/Tcirg1 (ATP6i)
src (tyrosine kinase)
CATK (collagenase)
TRAP (acid phosphatase)
αvβ3 (integrin)
Cln7 (chloride channel)
TRAF6 (cytoplasmic factor)

Osteoporosis: definition

- bone disease that develops when **bone mineral density** and bone mass decreases, or when the quality or structure of bone changes
- This can lead to a decrease in **bone strength** that can increase the risk of **fractures**

- Fragility fracture hip, spine without other metabolic bone diseases
- BMD (hip, spine, 1/3 distal radius) T-score ≤ -2.5
- BMD T-score between -1 and -2.5 with high FRAX score
- BMD T-score between -1 and -2.5 with fracture of humerus, pelvis, distal forearm

OP >>> fracture

OP = low bone strength >> Fracture

Bone strength = mass + quality

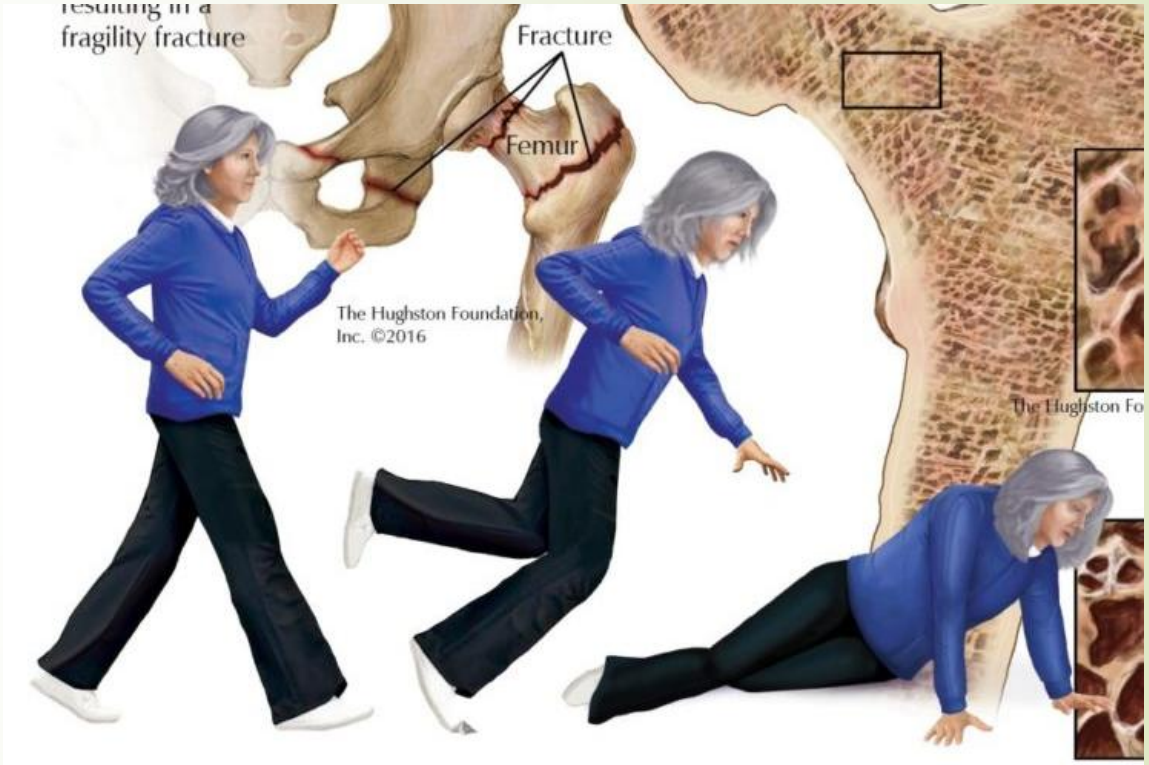
Fragility fracture, let's pick

- Film appearance
- Posture
- Age group
- Bone site
- Mechanism

Fragility fractures

How can you diagnose of fragility fracture

- **fractures that result from mechanical forces that would not ordinarily result in fracture**, known as low-level (or 'low energy') trauma. The World Health Organization (WHO) has quantified this as forces equivalent to a fall from a standing height or less



NICE UK Clinical guideline [CG146]

Osteoporosis: definition

- bone disease that develops when **bone mineral density** and bone mass decreases, or when the quality or structure of bone changes
- This can lead to a decrease in **bone strength** that can increase the risk of **fractures**

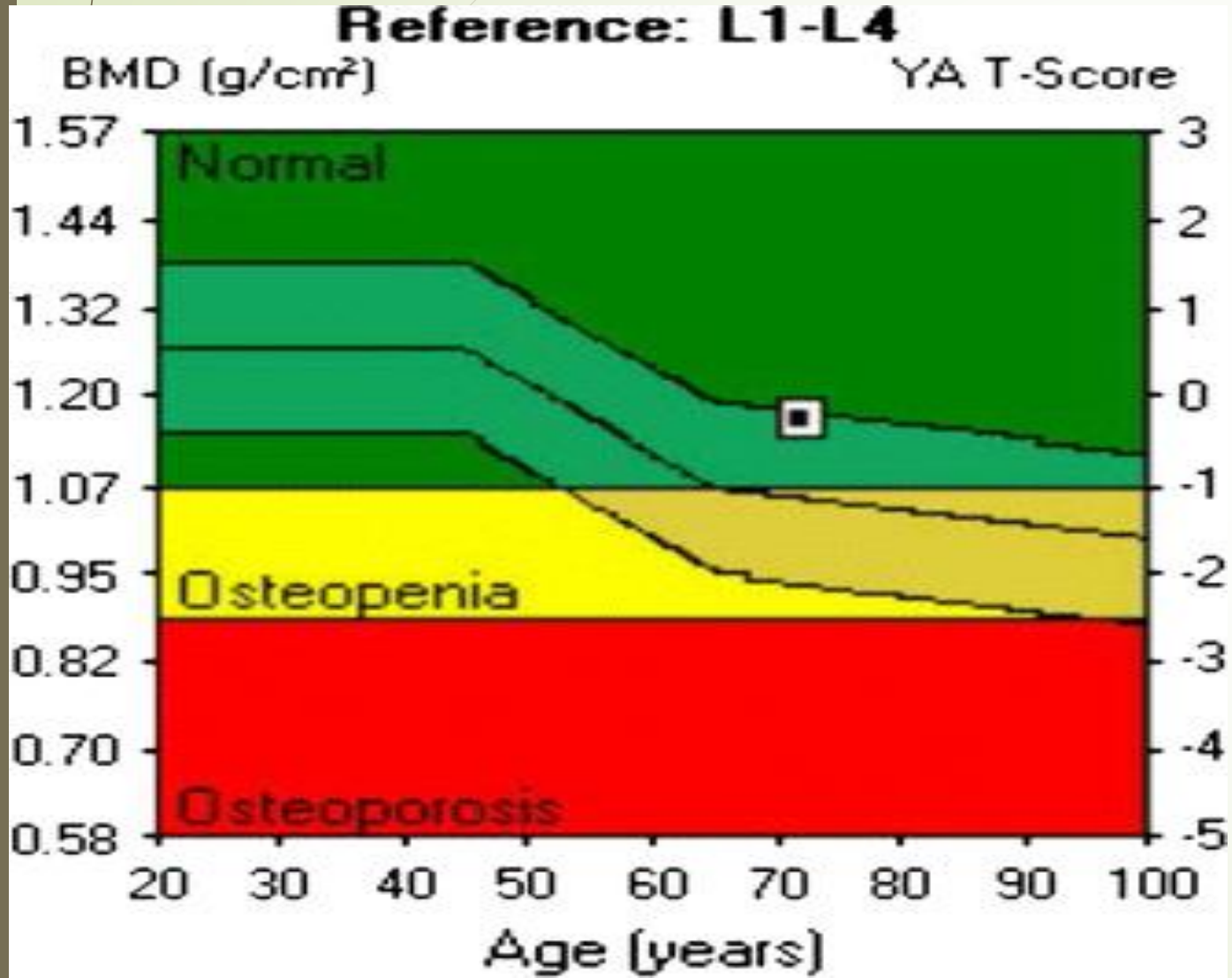
- Fragility fracture hip, spine without other metabolic bone diseases
- BMD (hip, spine, 1/3 distal radius) T-score ≤ -2.5
- BMD T-score between -1 and -2.5 with high FRAX score
- BMD T-score between -1 and -2.5 with fracture of humerus, pelvis, distal forearm

OP >>> fracture

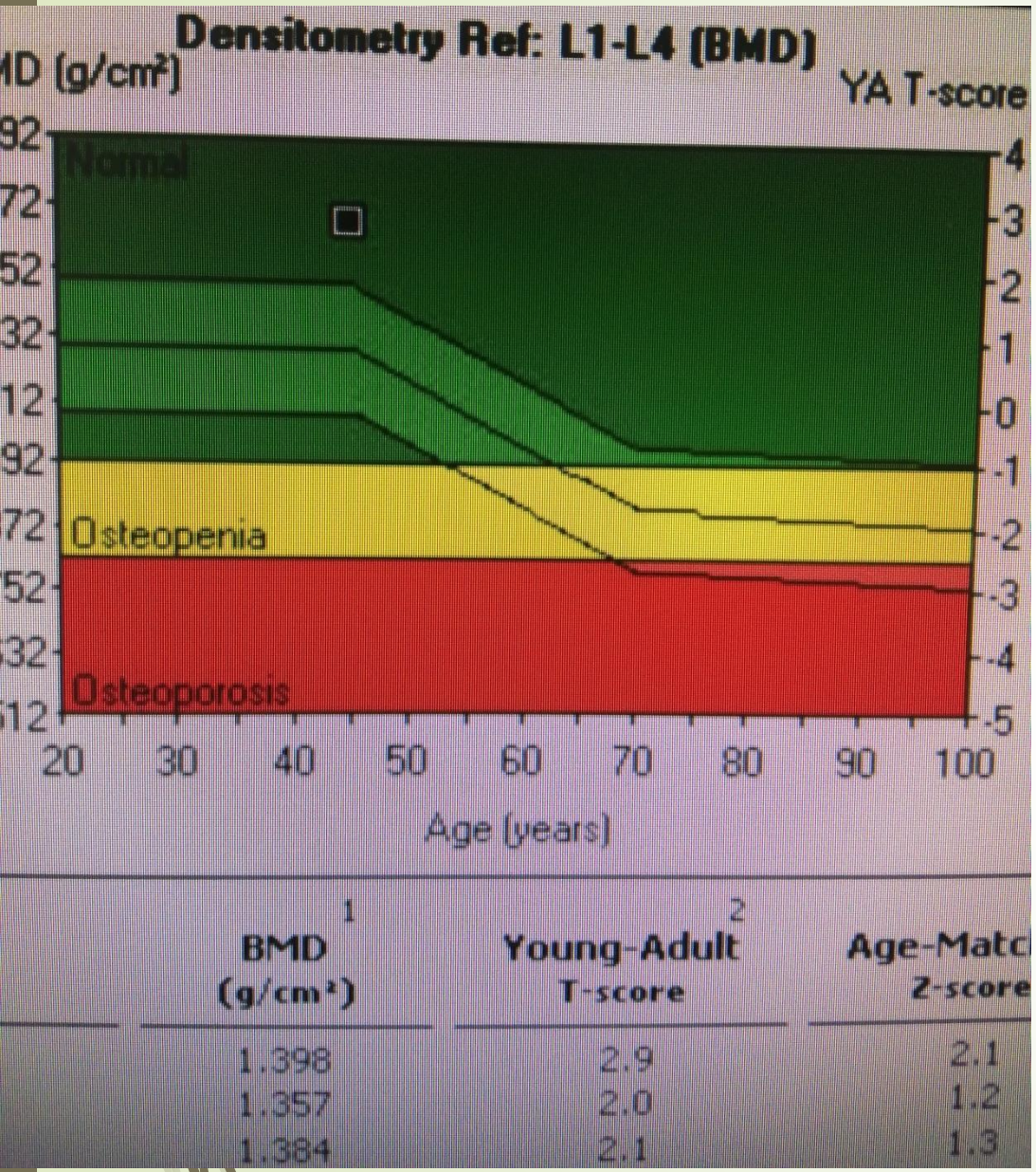
OP = low bone strength >> Fracture

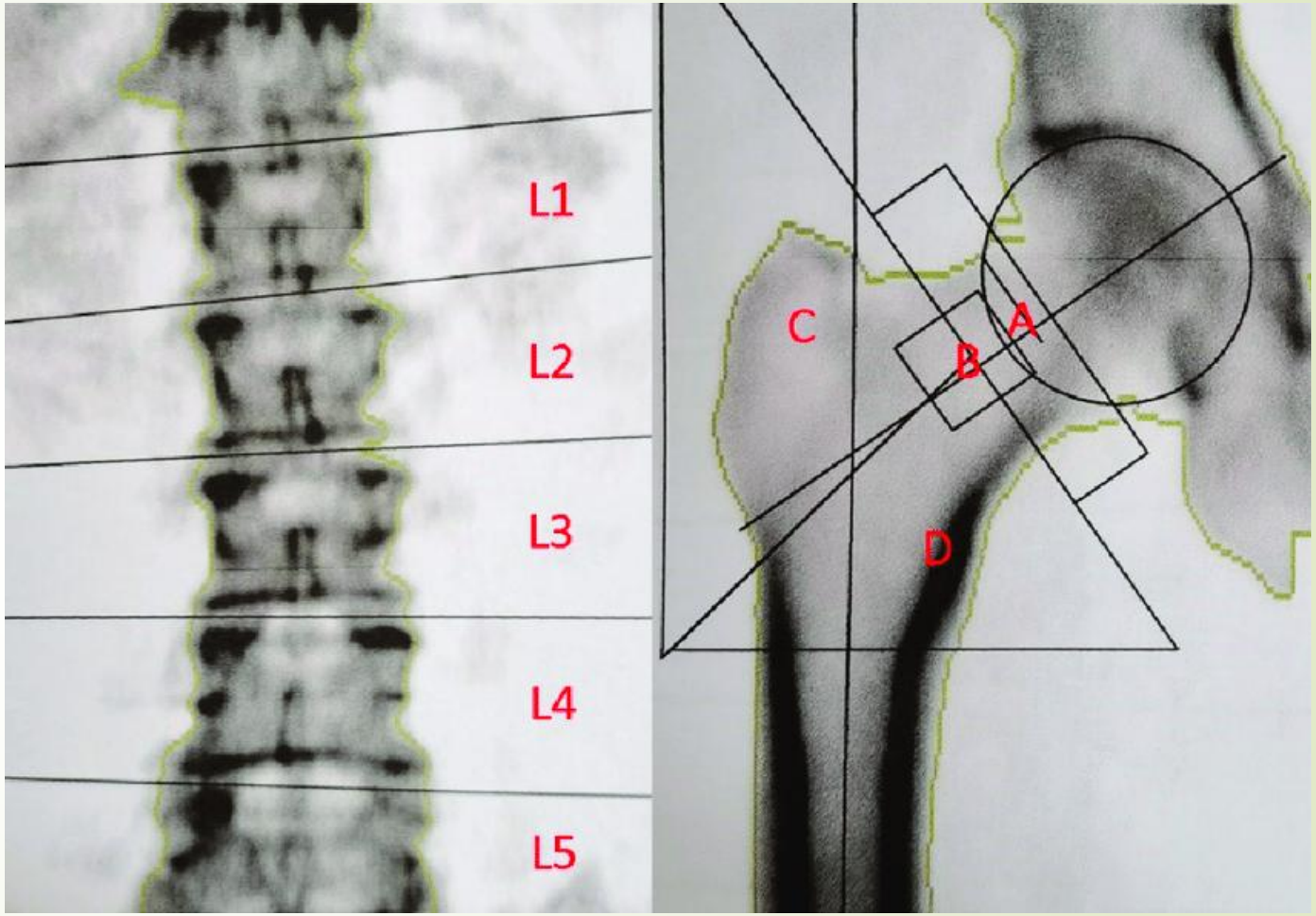
Bone strength = mass + quality

BMD

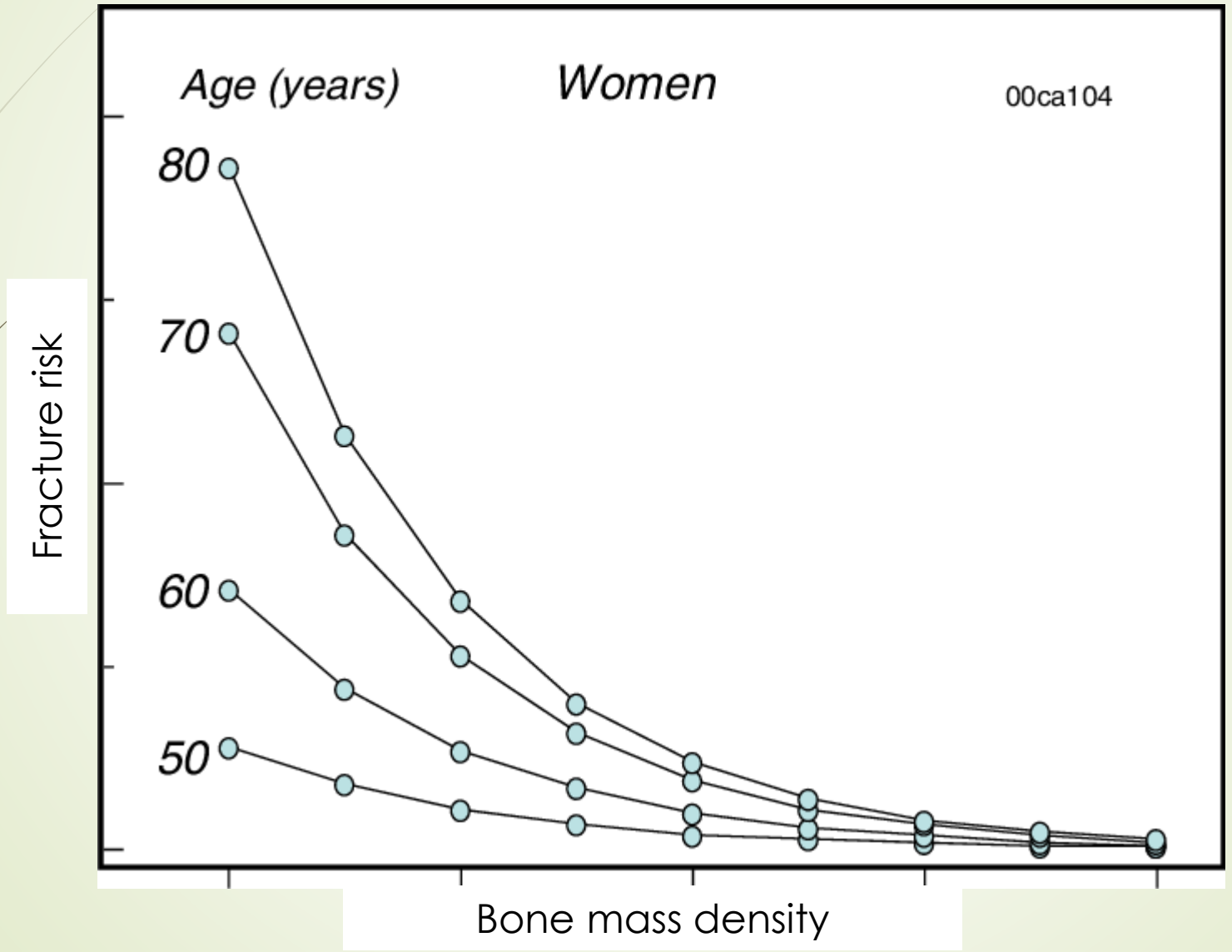


- Mineral amount per bone area
- g/cm2
- Same ethnic
- T-Score = standard deviation of young vs mean of young
- Z-Score = standard deviation of same age vs mean of same age


Densitometry Ref: L1-L4 (BMD)
MD (g/cm²)
YA T-score
92
72
52
32
12
92
72
52
32
12
4
3
2
1
0
-1
-2
-3
-4
-5
Normal
Osteopenia
Osteoporosis
20
30
40
50
60
70
80
90
100
Age (years)
BMD [1] (g/cm²)
Young-Adult [2] T-score
Age-Matc Z-score
1.398
1.357
1.384
2.9
2.0
2.1
2.1
1.2
1.3


L1
L2
L3
L4
L5
C
A
B
D

BMD is not that important



Osteoporosis: definition

- bone disease that develops when bone mineral density and bone mass decreases, or whe quality or structure of bone ch
- This can lead to a decrease in bone strength that can increase the risk of fractures

Fracture risk assessment tool (FRAX)

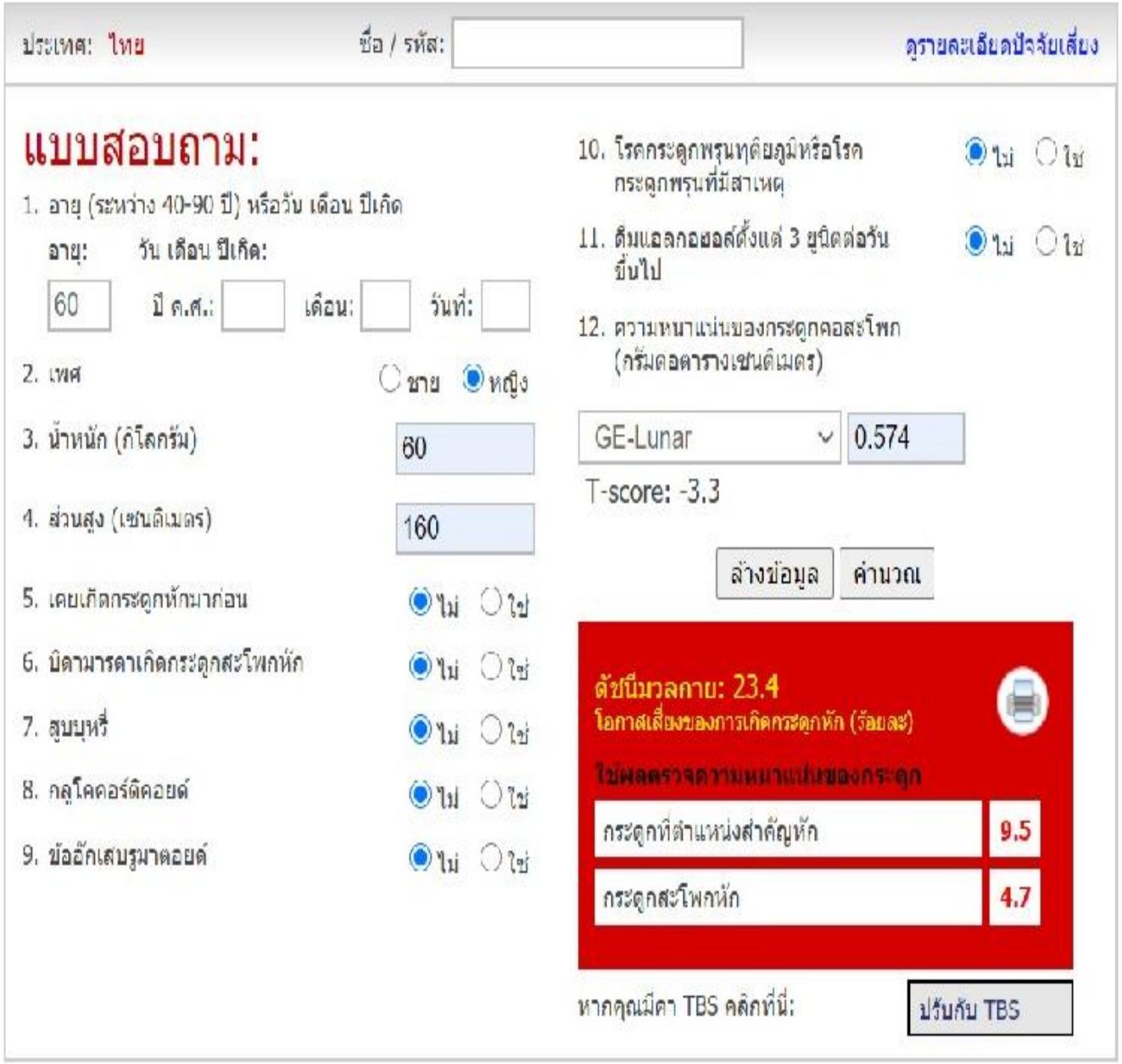
- Fragility fracture hip, spine without other metabolic bone diseases
- ne, 1/3 distal radius)
- BMD T-score between -1 and -2.5 with high FRAX score
- BMD T-score between -1 and -2.5 with fracture of humerus, pelvis, distal forearm

OP >>> fracture

OP = low bone strength >> Fracture

Bone strength = mass + quality

ตอบคำถามข้างล่างเพื่อประเมินโอกาสเสี่ยงของการเกิดกระดูกหักที่ระยะเวลา 10 ปี





การแปลงน้ำหนัก

ปอนด์ ➡ กิโลกรัม

การแปลงค่า

การแปลงส่วนสูง

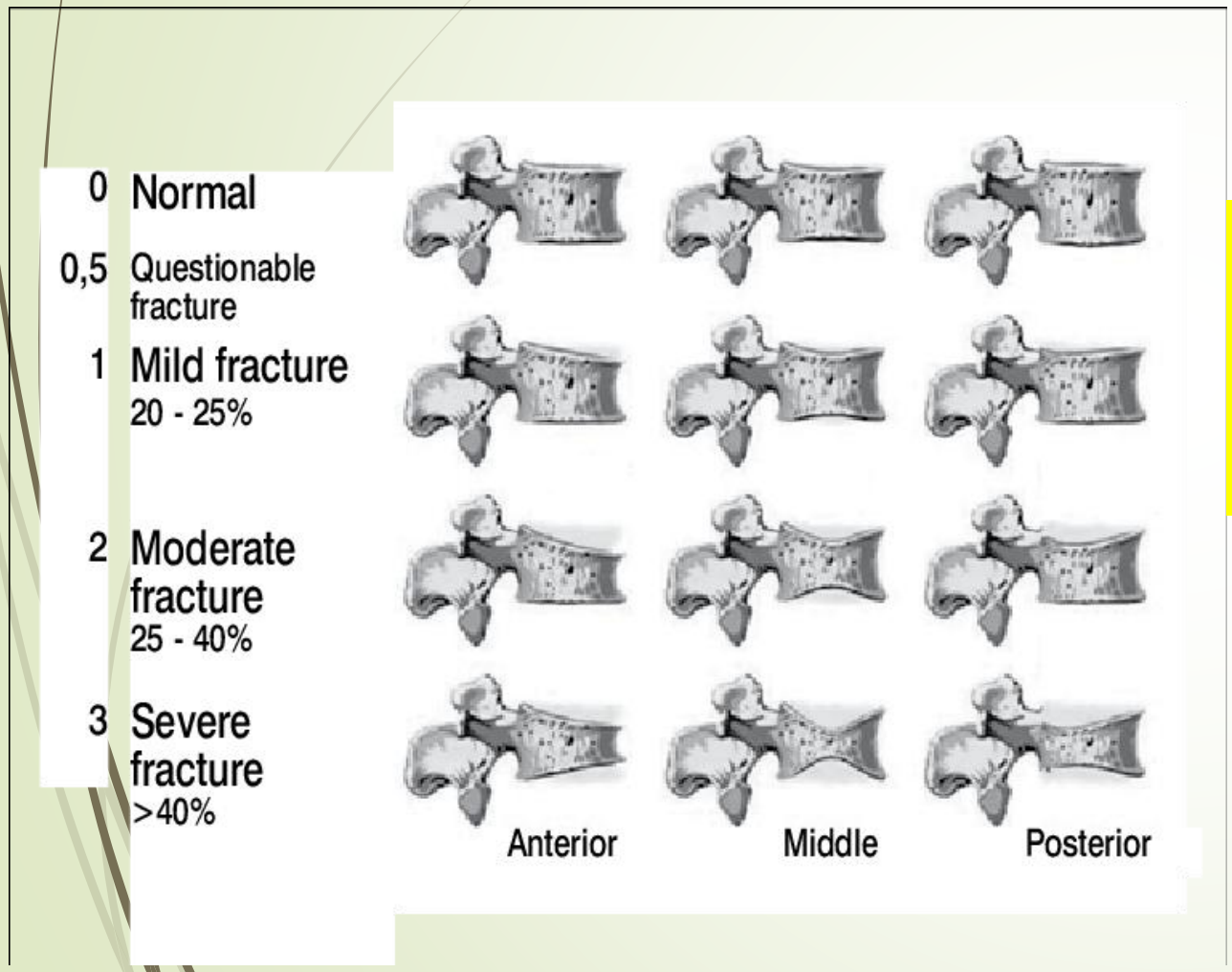
นิ้ว ➡ เซนติเมตร

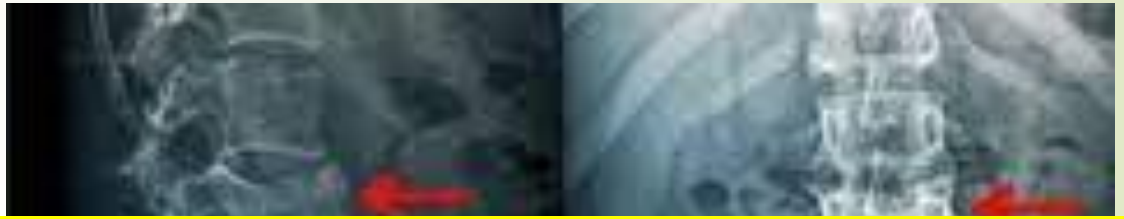
การแปลงค่า

00133425

จำนวนผู้ที่ใช้บริการประเมินความเสี่ยงของการเกิดกระดูกหักตั้งแต่วันที่ 1 มิถุนายน ค.ศ. 2011

Prevalent fracture





Fragility spine fracture
- Half is known
- Common site = lower T, upper L
- Film TL spine is indicated



Fracture risk vs BMD

- Factors

falls

Stroke
Weak, unfit
Unsteady gait
Drunk, drugs
Poor agility
Vision

Bone quality

cushion

Bone turnover
TBS
Collagen
Trabecular number
Bone shape

Table 12
Causes of Secondary Osteoporosis in Adults[a]

Endocrine or metabolic causes	Nutritional/ GI conditions	Drugs	Disorders of collagen metabolism	Other
Acromegaly	Alcoholism	Anti-epileptic drugs[b]	Ehlers-Danlos syndrome	AIDS/HIV
Diabetes mellitus	Anorexia nervosa	Aromatase inhibitors	Homocystinuria due to cystathionine deficiency	Ankylosing spondylitis
Type 1	Calcium deficiency	Chemotherapy/ immunosuppressants	Marfan syndrome	Chronic obstructive pulmonary disease
Type 2	Chronic liver disease	Medroxyprogesterone acetate	Osteogenesis imperfecta	Gaucher disease
Growth hormone deficiency	Malabsorption syndromes/ malnutrition (including celiac disease, cystic fibrosis, Crohn disease, and gastric resection or bypass)	Glucocorticoids		Hemophilia
Hypercortisolism		Gonadotropin-releasing hormone agents		Hypercalciuria
Hyperparathyroidism		Heparin		Immobilization
Hyperthyroidism		Lithium		Major depression
Hypogonadism		Proton pump inhibitors		Myeloma and some cancers
Hypophosphatasia		Selective serotonin- reuptake inhibitors		Organ transplantation
Porphyria		SGLT2-inhibitors		Renal insufficiency/ failure
Pregnancy	Total parenteral nutrition	Thiazolidinediones		Renal tubular acidosis
	Vitamin D deficiency	Thyroid hormone (in supraphysiologic doses)		Rheumatoid arthritis
				Systemic mastocytosis
				Thalassemia

Further investigations include….

Table 12
Causes of Secondary Osteoporosis in Adults[a]

Endocrine or metabolic causes	Nutritional/ GI conditions	Drugs	Disorders of collagen metabolism	Other
Acromegaly	Alcoholism	Anti-epileptic drugs[b]	Ehlers-Danlos syndrome	AIDS/HIV
Diabetes mellitus	Anorexia nervosa	Aromatase inhibitors	Homocystinuria due to cystathionine deficiency	Ankylosing spondylitis
Type 1	Calcium deficiency	Chemotherapy/ immunosuppressants	Marfan syndrome	Chronic obstructive pulmonary disease
Type 2	Chronic liver disease	Medroxyprogesterone acetate	Osteogenesis imperfecta	Gaucher disease
Growth hormone deficiency	Malabsorption syndromes/ malnutrition (including celiac disease, cystic fibrosis, Crohn disease, and gastric resection or bypass)	Glucocorticoids		Hemophilia
Hypercortisolism	Total parenteral nutrition	Gonadotropin-releasing hormone agents		Hypercalciuria
Hyperparathyroidism	Vitamin D deficiency	Heparin		Immobilization
Hyperthyroidism		Lithium		Major depression
Hypogonadism		Proton pump inhibitors		Myeloma and some cancers
Hypophosphatasia		Selective serotonin- reuptake inhibitors		Organ transplantation
Porphyria		SGLT2-inhibitors		Renal insufficiency/ failure
Pregnancy		Thiazolidinediones		Renal tubular acidosis
		Thyroid hormone (in supraphysiologic doses)		Rheumatoid arthritis
				Systemic mastocytosis
				Thalassemia

- Hx
- PE
- Laboratory

Metabolic bone diseases

Mineralization: OP vs OM

- Hypocalcemia
  - Vitamin D deficiency/resistance
  - Absorption/severe malnutrition
- Hypophosphatemia
  - XLH
  - TIO
  - Hyperparathyroidism
  - Hypomagnesemia
- Hypophosphatasia

OB, OC, matrix

- Osteogenesis imperfecta
- Osteopetrosis
- Pycnodysostosis
- Paget
- Tumor bone diseases

A 61-year-old man broke lt hip from a simple fall



- DM for 6 years, OAD
- 4 months: Adenocarcinoma of rectum CCRT
- 2 months: Partial hepatectomy from liver mass: granulomatous lesion and found renal deteriorate
- Last week simple falls leading to left hip pain

2ry osteoporosis

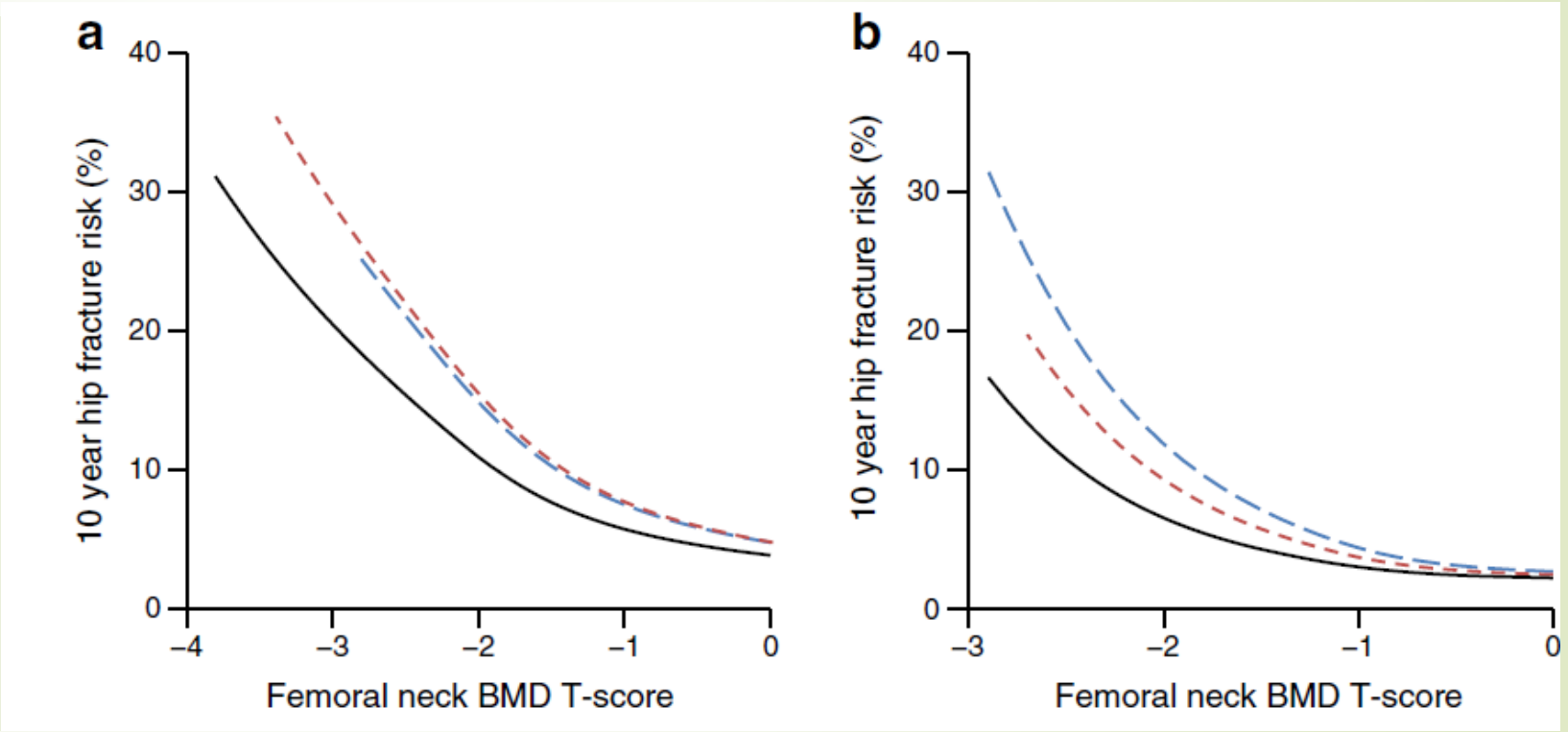
- CBC
- Cr
- Electrolyte
- Ca, PO, ALP
- TFT
- PTH
- Vitamin D

- MM
- CKD, RTA
- Hyperparathyroidism
- Hyperthyroidism
- Hypovitaminosis D*
- Cushing, hypogonad, gaucher, OI, mastocytosis, as clinical indicates

Type 2 DM

A women, B men
Solid = nonDM
Dash blue = insulin
Dash red = noninsulin

- Poorer bone quality, more fracture
  - As if bone is very old, add 10 years, or
  - As if they are rheumatoid, or
  - As if BMD is poorer, T score -0.5



Diabetologia (2017) 60:1170–1179

Osteoporosis: definition

- bone disease that develops when **bone mineral density** and bone mass decreases, or when the quality or structure of bone changes
- This can lead to a decrease in **bone strength** that can increase the risk of **fractures**

- Fragility fracture hip, spine without other metabolic bone diseases
- BMD (hip, spine, 1/3 distal radius) T-score ≤ -2.5
- BMD T-score between -1 and -2.5 with high FRAX score
- BMD T-score between -1 and -2.5 with fracture of humerus, pelvis, distal forearm

OP >>> fracture

OP = low bone strength >> Fracture

Bone strength = mass + quality

Metabolic bone diseases

ENDOCRINE PSU

Mineralization: OP vs OM

- Hypocalcemia
  - Vitamin D deficiency/resistance
  - Absorption/severe malnutrition
- Hypophosphatemia
  - XLH
  - TIO
  - Hyperparathyroidism
  - Hypomagnesemia
- Hypophosphatasia

OB, OC, matrix

- Osteogenesis imperfecta
- Osteopetrosis
- Pycnodysostosis
- Paget
- Tumor bone diseases

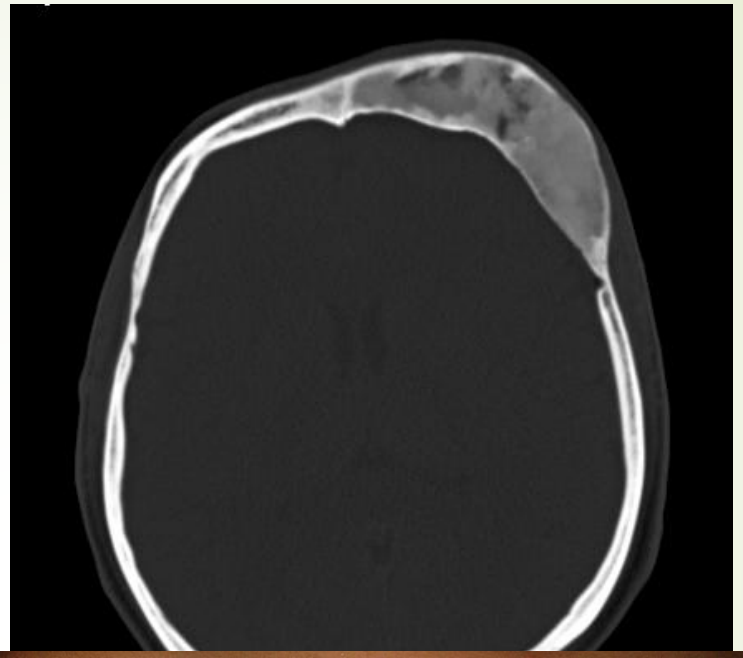




ENDOCRINE PSU

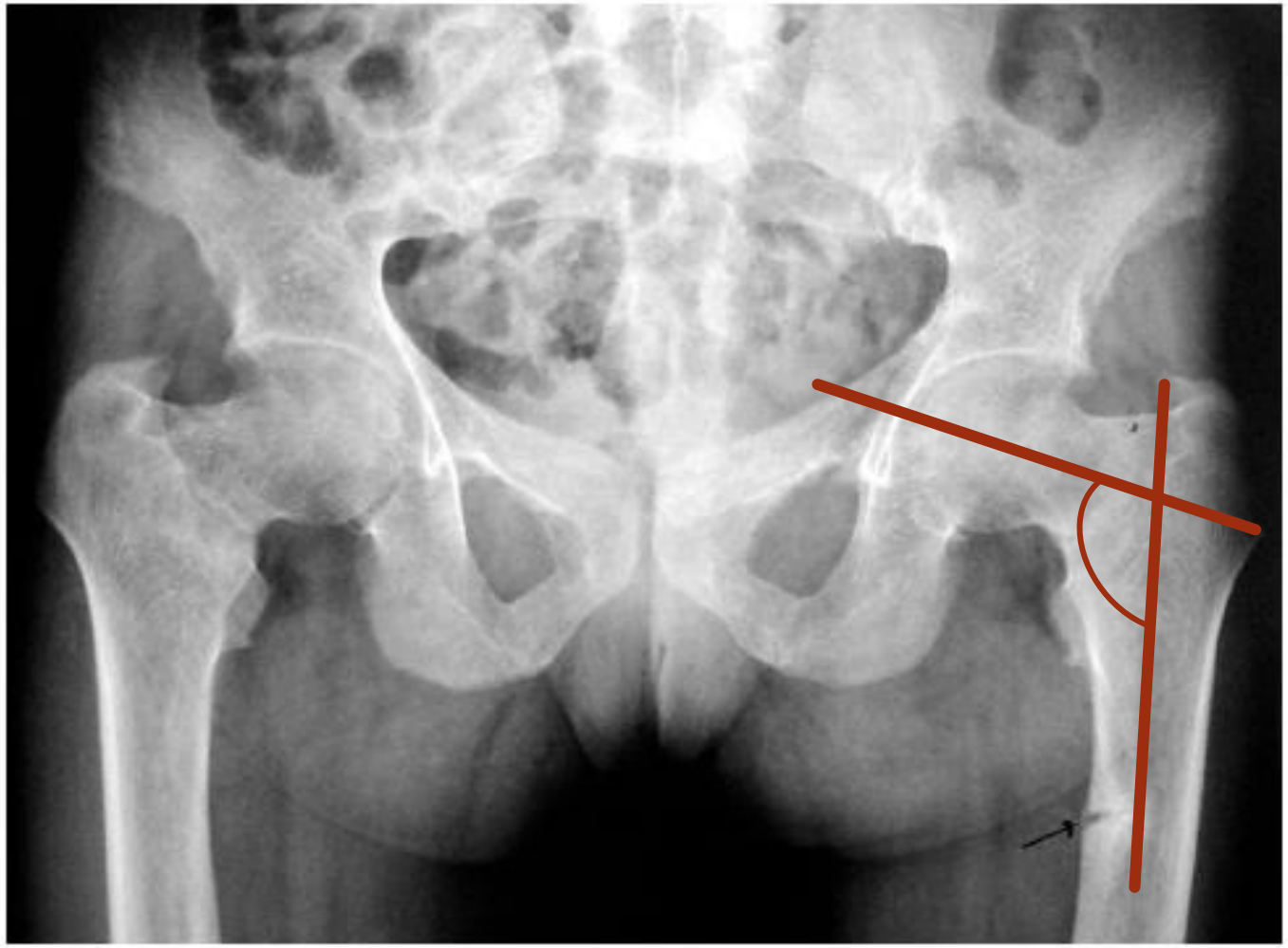
Fibrous dysplasia



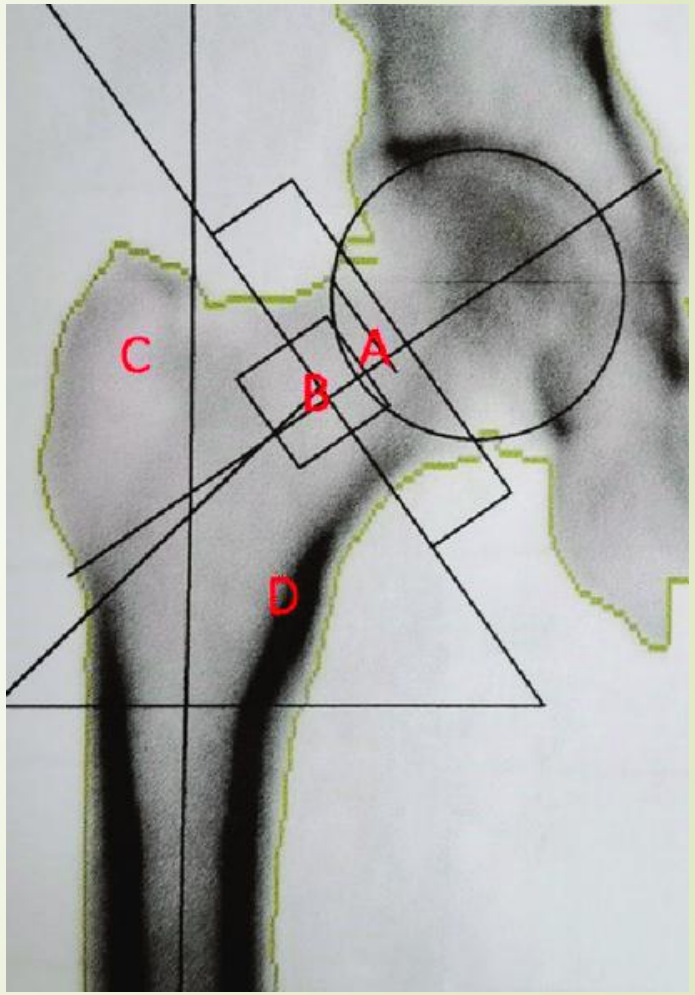




Café au lait: coast of Maine



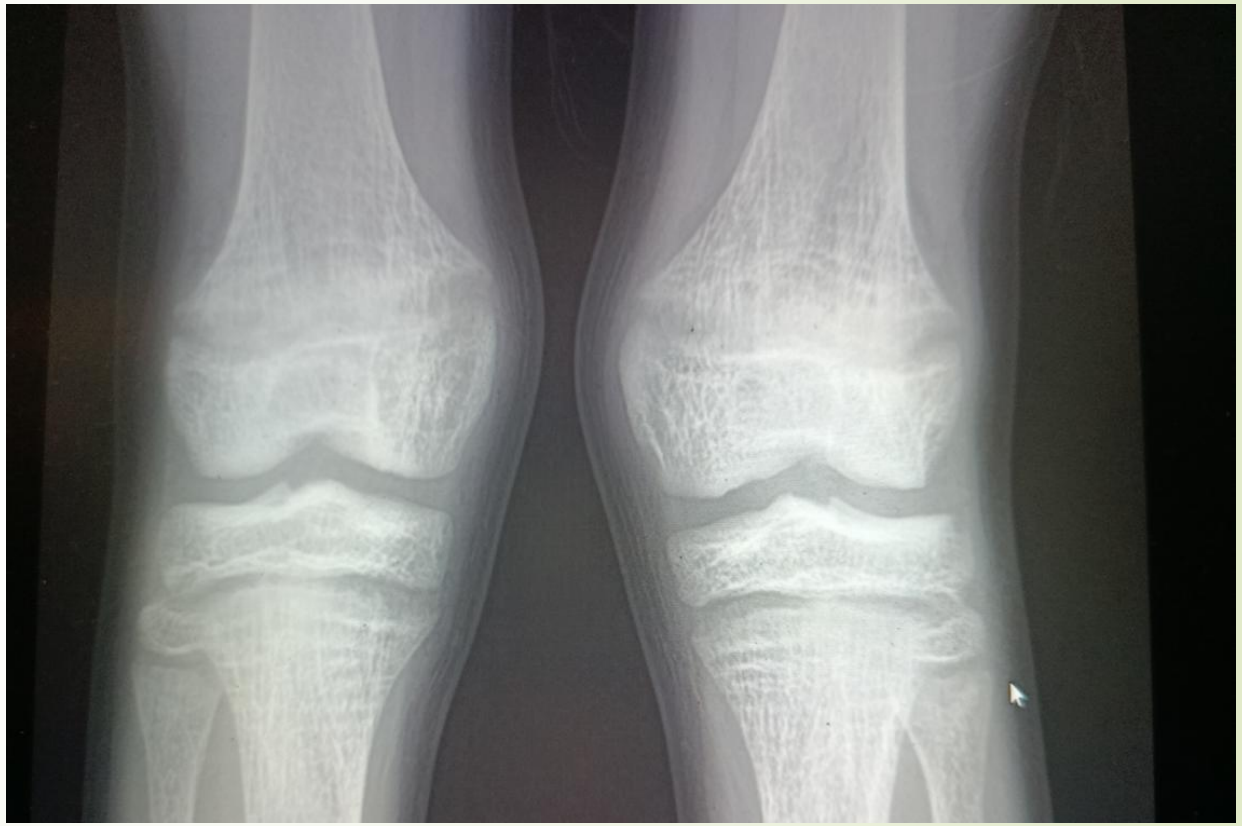
Angle 120-140
<120 = coxa vara





Fraying

Flaring



Looser zone
Incomplete fracture

ENDOCRINE PSU







- Expansile thick cortex
- Sclertic bone
- Coarse trabeculation

ENDOCRINE PSU

Table 12
Causes of Secondary Osteoporosis in Adults[a]

Endocrine or metabolic causes	Nutritional/ GI conditions	Drugs	Disorders of collagen metabolism	Other
Acromegaly Diabetes mellitus Type 1 Type 2 Growth hormone deficiency Hypercortisolism Hyperparathyroidism Hyperthyroidism Hypogonadism Hypophosphatasia Porphyria Pregnancy	Alcoholism Anorexia nervosa Calcium deficiency Chronic liver disease Malabsorption syndromes/ malnutrition (including celiac disease, cystic fibrosis, Crohn disease, and gastric resection or bypass) Total parenteral nutrition Vitamin D deficiency	Anti-epileptic drugs[b] Aromatase inhibitors Chemotherapy/ immunosuppressants Medroxyprogesterone acetate Glucocorticoids Gonadotropin-releasing hormone agents Heparin Lithium Proton pump inhibitors Selective serotonin- reuptake inhibitors SGLT2-inhibitors Thiazolidinediones Thyroid hormone (in supraphysiologic doses)	Ehlers-Danlos syndrome Homocystinuria due to cystathionine deficiency Marfan syndrome Osteogenesis imperfecta	AIDS/HIV Ankylosing spondylitis Chronic obstructive pulmonary disease Gaucher disease Hemophilia Hypercalciuria Immobilization Major depression Myeloma and some cancers Organ transplantation Renal insufficiency/ failure Renal tubular acidosis Rheumatoid arthritis Systemic mastocytosis Thalassemia


LearningRadiology.com (C)
All Rights Reserved



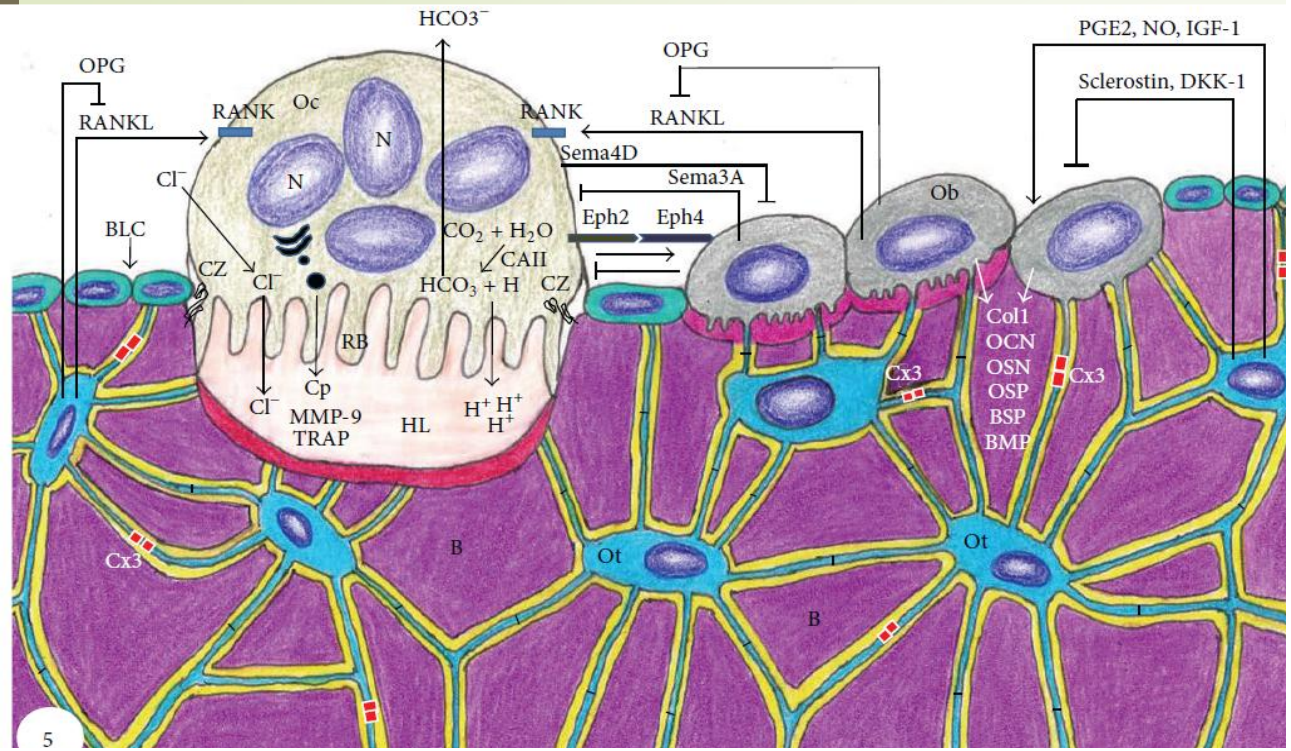


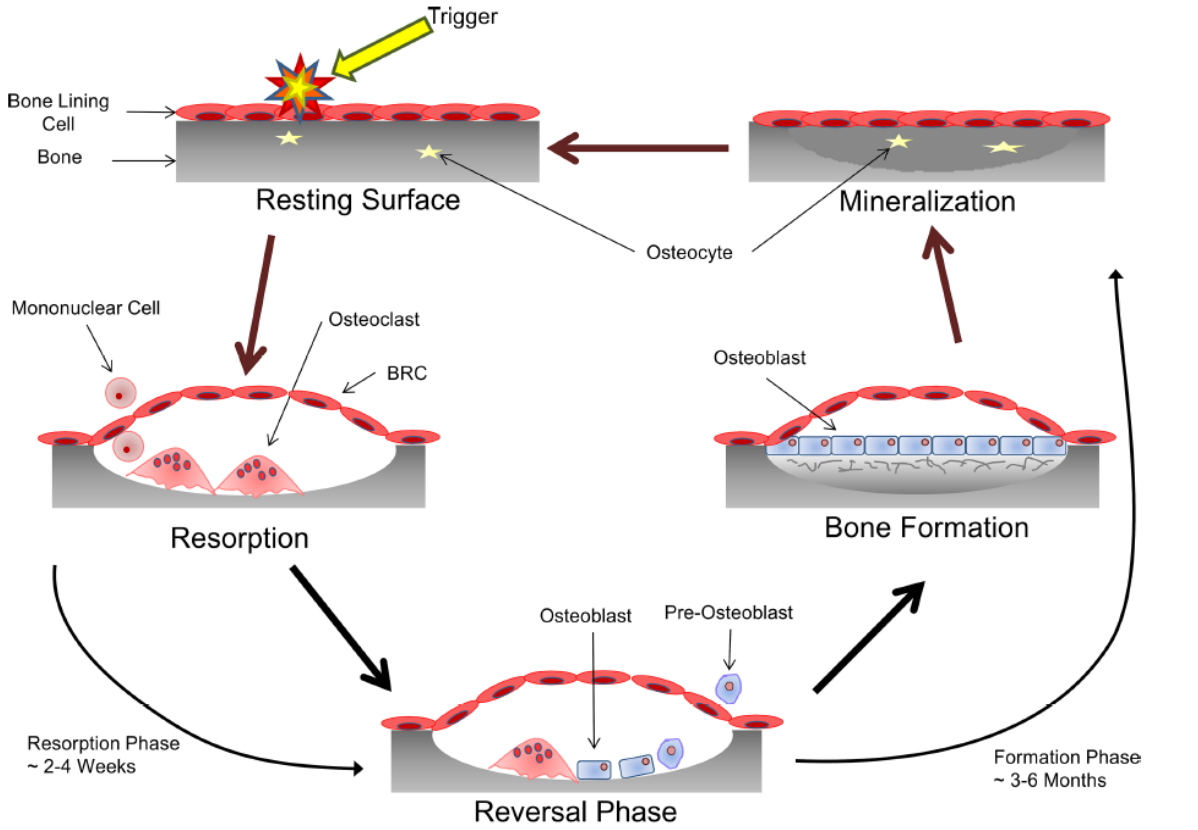
-Osteitis fibrosa cystiga
-Brown tumor
-Sclerotic bone

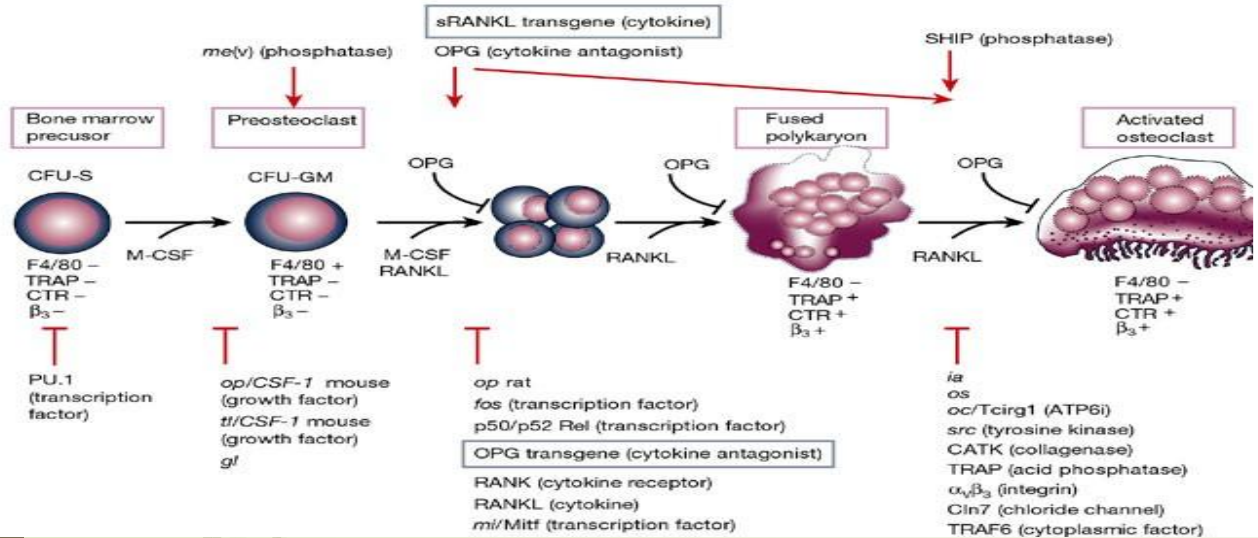


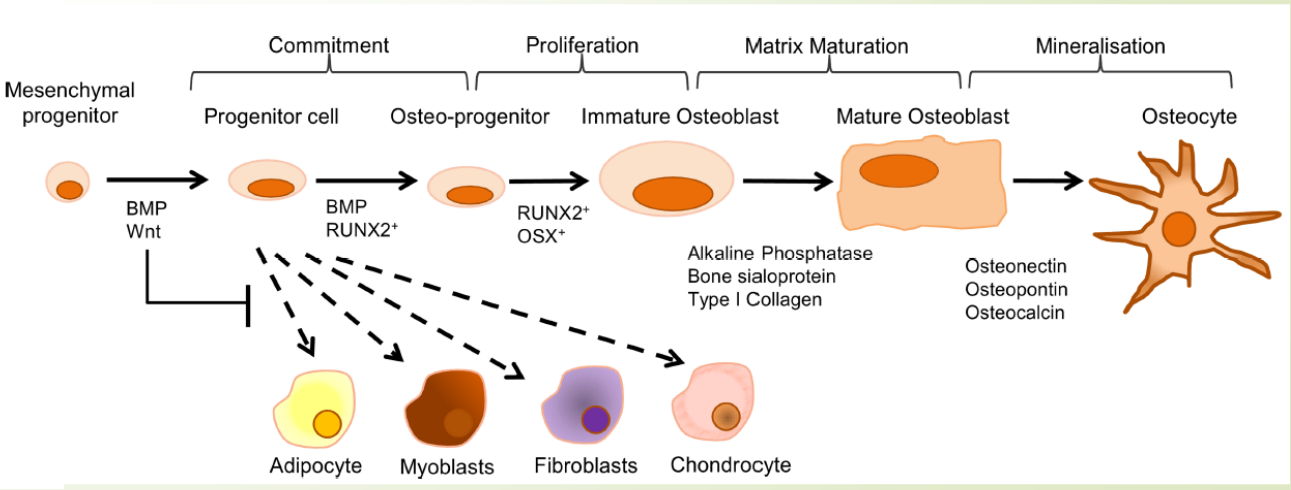


Therapy of osteoporosis









Antiosteoporosis medication

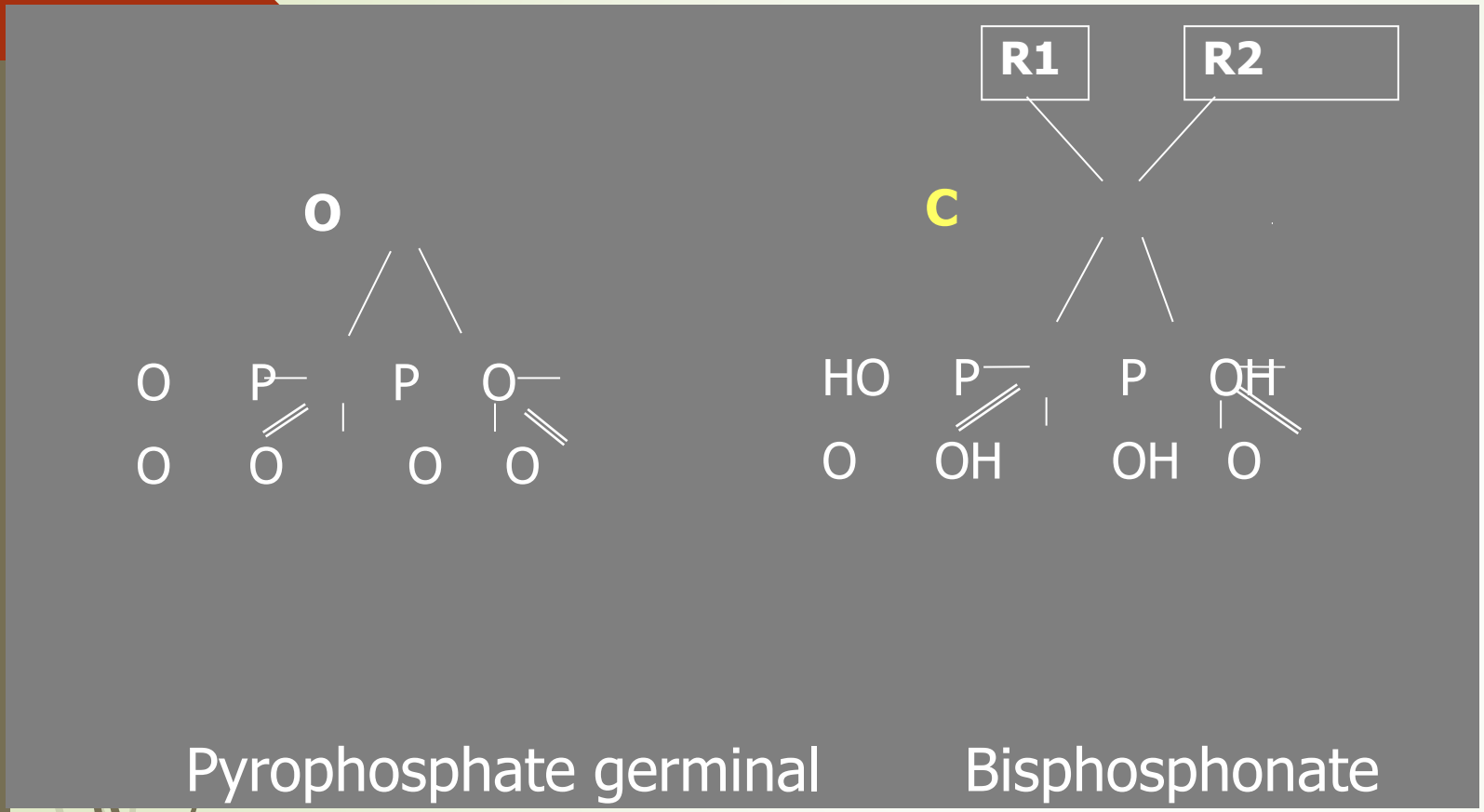
Aborted medication

- Estrogen
- Fluoride
- Odanacatib: Cathepcin K inhibitor
- Strontium ranelate

Available medication

- Bisphosphonate
- antiTNF (denosumab)
- SERMs (raloxifene)
- Calcitonin
- PTH (teriparatide)
- Abaloparatide
- Romosozumab

bisphosphonate



R1

- Binding to calcium
- OH group

R2

- N: biological
  - Simple molecule
  - More substitute
  - Heterocyclic ring
- Binding to hydroxyapatite [+charge on bone surface]

retention in skeleton

resolution from skeleton

reuptake to skeleton

mineral property & cellular effect

Bisphosphonate: mechanism of action

- Physicochemical
  - Bind strongly to crystal
  - Inhibit formation of crystal
  - Inhibit aggregation of crystal
  - Inhibit dissolution of crystal

BP biological & physical property

- Binding to hydroxyapatite crystal

  retention in skeleton

  resolution from skeleton

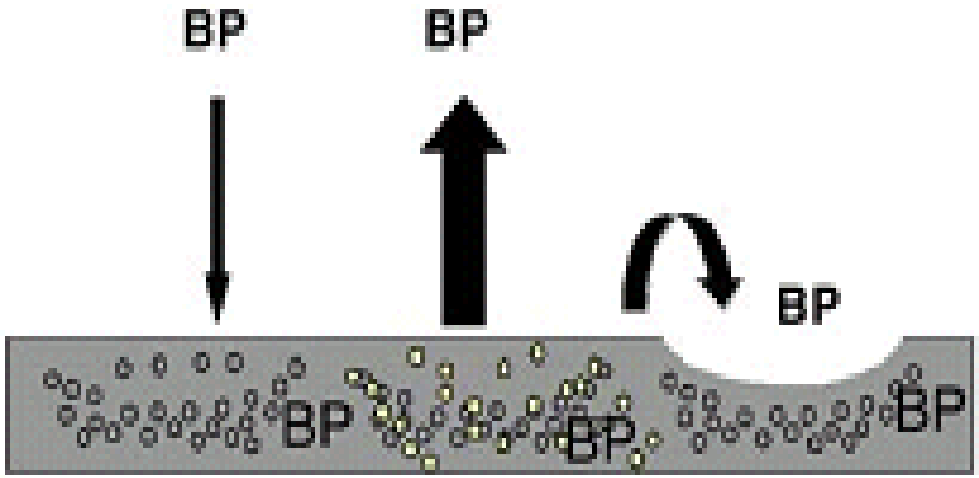
  reuptake to skeleton

  mineral property & cellular effect

Bisphosphonate Uptake and Detachment from Bone Surfaces

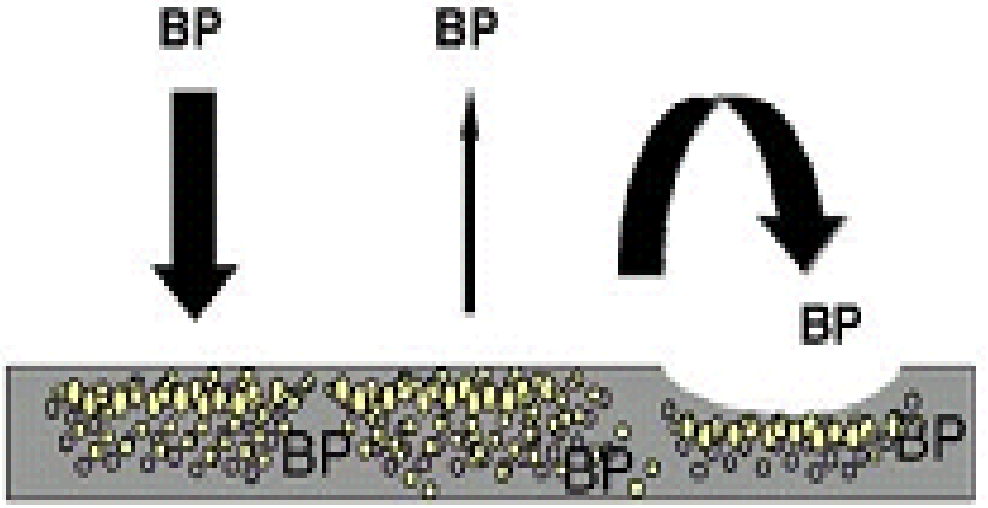
Lower Affinity BP

- Weaker uptake
- Higher desorption
- Lower re-attachment
- More diffusion in bone



Higher Affinity BP

- Avid uptake
- Lower desorption
- Higher re-attachment
- Less diffusion in bone



osteoporosis

- Bone is dynamic in order to keep normal metabolism of calcium, phosphorus, acid base, and keep normal growth, mobility, body posture and vital organ protection
- Osteoporosis is the most common metabolic bone disease causing poorer bone strength according to low bone mass and microarchitecture damage leading to fracture, rule out other conditions
- Dx of osteoporosis = fragility fracture, or hi fracture risk

